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Characterizing NGC 6383: A study of fundamental properties, pre-main sequence stars, mass segregation, and age using *Gaia* DR3 and 2MASS

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ABSTRACT

Context. This study focuses on the young open cluster NGC 6383, situated in the Carina-Sagittarius arm within the Sh 2-012 star-formation region. Previous studies have provided estimates of the distance, age, and structural properties of the cluster. The presence of pre-main sequence and low-mass stars, combined with *Gaia* DR3 and 2MASS data, makes NGC 6383 a compelling target for a refined characterization.
Aims. We aim to identify cluster members, determine fundamental parameters, assess mass segregation, and derive the age and distance of NGC 6383 from *Gaia* DR3 and 2MASS data.
Methods. We employed Bayesian analysis and machine-learning techniques, including the Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) for member identification, PyMC for the Bayesian modeling, the Sagitta neural network for the identification and age estimation of pre-main sequence stars, and ASteCA for isochrone fitting.
Results. We identified 321 likely cluster candidates with $\tilde{p} > 0.5$ and used a reference sample of 254 sources with $\tilde{p} \geq 0.6$ to derive the cluster parameters. The mode cluster age is $3.53^{+1.40}_{-1.00}$ Myr and the distance is 1.11 ± 0.06 kpc. The ~~fitted core and King outer radii are 1.95 ± 0.19 and 40.4 ± 14.3 arcmin, respectively, with the outer radius weakly constrained by the search-field size~~core radius is $1.96^{+0.19}_{-0.16}$ arcmin ($0.63^{+0.06}_{-0.05}$ pc); the King outer radius, measured on a 70 arcmin extraction wide enough to constrain the field background beyond the cluster, is 54^{+7}_{-11} arcmin ($17.4^{+2.3}_{-3.6}$ pc). Given the cluster's dynamical youth (age ~~$\simeq 0.12$~~ $\simeq 0.14$ of the half-mass relaxation time), the observed mass segregation among binary stars is consistent with a primordial or near-primordial origin. The color-magnitude diagram (CMD) shows a well-defined main sequence and a population of pre-main sequence stars, consistent with a recent, possibly extended star-formation episode spanning approximately 1.6 to 6.3 Myr in age. The software used for this investigation is released with the paper.
Conclusions. Combining machine-learning membership with Bayesian parameter estimation, our analysis provides updated parameters for NGC 6383, an apparent age spread consistent with recent, possibly extended star formation, and a segregation pattern consistent with a primordial origin or with rapid early-dynamical segregation (near-primordial).

Key words. techniques: photometric – parallaxes – proper motions – stars: distances – stars: pre-main sequence – open clusters and associations: individual: NGC 6383

1. Introduction

Membership determination underpins any star-cluster characterization: distances, ages, and structural parameters are only as reliable as the member list they rest on. Methods have evolved from early spectroscopic and photometric studies (e.g. Trumpler 1930; Sanford 1949; Lindoff 1968; Fitzgerald et al. 1978) to multi-parameter membership analyses (Pandey et al. 1989; Battinelli & Capuzzo-Dolcetta 1991; Kharchenko et al. 2005; Paunzen et al. 2007; Aidelman et al. 2018), whose differing assumptions can return discrepant member lists for the same cluster. Bayesian and machine-learning approaches provide a complementary route that makes the selection criteria explicit and reproducible.

NGC 6383[1] is a young open cluster located in the Carina-Sagittarius arm, within the Sh 2-012 star-formation region (Sharpless 1959). It is commonly associated with the broader Sgr OB1 star-forming complex, together with the open clusters NGC 6530 and NGC 6531 (Rauw & De Becker 2008). The Galactic coordinates are $\ell = 355.68°$ and $b = 0.05°$ (Conrad et al. 2017). The distance to the cluster has been estimated by various authors, ranging from an upper limit of 2.13 kpc (Trumpler 1930) to lower limits of 0.760 kpc (Sanford 1949) and 0.840 kpc (Aidelman et al. 2018). The cluster spans an angular size of 20.0 arcmin (Morales et al. 2013) and is estimated to be between 1.70 and 5.00 Myr old (Fitzgerald et al. 1978; Pandey et al. 1989; Kharchenko et al. 2005; Paunzen et al. 2007), though Lindoff (1968) ~~estimated an age of 20.0 Myr~~and Battinelli & Capuzzo-Dolcetta (1991) estimated ages of ~ 20 Myr. The reddening $E(B-V) = 0.320 \pm 0.020$ derived by Rauw & De Becker (2008) aligns with estimates from previous authors (Becker & Fenkart 1971; Fitzgerald et al. 1978; Lloyd Evans 1978; The et al. 1985; Pandey et al. 1989; Feinstein 1994; Paunzen et al. 2007), but dif-

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[1] NGC 6383, also known as NGC 6374 in the New General Catalog (Rauw & De Becker 2008) and classified as Collinder 334 and Collinder 335 in the Collinder catalog, was initially misclassified in the original Collinder catalog (Collinder 1931).

fers from Aidelman et al. (2018), who determined a reddening of $E(B-V) = 0.51 \pm 0.03$. This reddening is due to a surrounding large shell-structured ionized HII region with a radius of approximately 1.00 deg.

HD 159176, a double-lined spectroscopic binary composed of O7V-type stars located in the projected center of the cluster, is responsible for ionizing the HII region (Penny et al. 2016). The estimated age of HD 159176 is ~~2.30~~2.3–~~2.80 Myr~~ 2.7 Myr (Rauw et al. 2010), in agreement with the earlier 2.8 ± 0.5 Myr of Fitzgerald et al. (1978). Rauw et al. (2010) found that the age of the most likely low-mass PMS candidates, 2–3 Myr, is in reasonable agreement with the estimated age of HD 159176, consistent with a common star-formation event if the binary is assumed to be a cluster member. Furthermore, Fitzgerald et al. (1978) proposed that HD 159176 initiated star formation in the cluster's core and beyond.

In a study by Aidelman et al. (2018), it was hypothesized that if the star labeled *Star No. 6* has begun to evolve and if HD 159176 is a blue straggler, the age of the cluster should be between 6.00 and 10.0 Myr. We treat this interpretation with caution because it depends on the membership and evolutionary status of HD 159176; spectroscopic studies classify the system as an O-type main-sequence binary (Stickland et al. 1993; Linder et al. 2007; Penny et al. 2016). The reported parameters span wide ranges, reflecting the diversity of the methods used to derive them. The cluster hosts sources that have recently entered the main-sequence track, while pre-main sequence stars (PMS) are also present (Kalari 2019), leading to considerable debate regarding the cluster's age. A summary of the historical results can be seen in Table A.1.

Relative to the *Gaia*-era survey catalogs (Cantat-Gaudin et al. 2020; He et al. 2022; Hunt & Reffert 2024), which provide membership and bulk parameters, and to the Hα/CTTS studies (Rauw et al. 2010; Kalari 2019), this work adds a quantified mass-segregation analysis including binaries, a PMS census combining Sagitta with a YSO fraction and an Hα cross-match, an explicit search-window sensitivity study of the membership and structural parameters, and a released, reusable pipeline.

This paper is structured as follows: Sect. ~~3 outlines the methodology, covering data acquisition and processing methods, including the use of the~~ 2 describes the *Gaia* DR3 and 2MASS ~~catalogs, and describes the membership determination using HDBSCAN and the Bayesian analysis for parameter estimation~~data and their preprocessing. Sect. ~~3.3 presents the results, including the determination of the cluster's age, distance, and structural properties,~~ 3 presents the analysis pipeline, the ~~analysis of~~ HDBSCAN membership determination, and the resulting member catalog. Sections 4 and 5 give the astrometric parameters (parallax, distance, proper motions, radial velocities, and center) and the cluster structure, each derived immediately after the corresponding method is described. Sect. 6 covers the isochrone-based age, extinction, and metallicity, and the pre-main-sequence content. Sect. 7 presents the luminosity function, stellar masses, mass segregation, and the ~~color-magnitude diagram. It also discusses the findings in the context of~~ dynamical state. Sect. 8 compares our results with previous studies, ~~addressing peculiar stars and comparisons with historical data. Finally,~~ and Sect. 9 ~~concludes the study, summarizing the key results and their implications~~summarizes the conclusions. The photometric system adopted includes the *Gaia* DR3 photometric bands ($G, G_{\rm BP}, G_{\rm RP}$) and the 2MASS bands ($J, H, K_s$).

2. ~~Methodology~~Data

We acquired data from the *Gaia* third Data Release (DR3) (Gaia Collaboration et al. 2016, 2023), executing a cone search of 40.0 arcmin radius centered on $(\alpha, \delta) = (263.6715^\circ, -32.5773^\circ)$ (ICRS) in the *Gaia* archive, which yielded 23740 sources. The query restricted the selection to parallaxes between 0.750 and 1.10 mas, based on the limits established in Pulgar-Escobar & Henríquez-Salgado (2024).

To enhance data reliability, we applied the astrometric fidelity parameter from Rybizki et al. (2022). This parameter, derived from a neural network analysis of 17 *Gaia* catalog metrics, assesses the trustworthiness of the astrometric solution. Sources with an astrometric fidelity above 0.5 were retained, narrowing the selection to 15480 sources, which represent 65% of the initial dataset.

Subsequent refinement ensured the inclusion of only sources for which all parameters $(\alpha, \delta, \ell, b)$, proper motions $(\mu_{\alpha*}, \mu_\delta)$, parallax (ϖ), and magnitudes $(G, G_{\rm BP}, G_{\rm RP})$ were available, reducing the dataset to 15276 sources. Systematic parallax offsets identified in Lindegren et al. (2021a) were corrected using the GAIADR3_ZEROPOINT package. Moreover, to address the bias in proper motion for bright sources ($G < 13$ mag), as discussed in Cantat-Gaudin & Brandt (2021), we applied a magnitude-based correction for sources with $G = 11$–13 mag, compensating for up to 80 μas yr^{-1} discrepancy between the frames of reference for bright and faint sources.

We also obtained the J, H, and K_s magnitudes from the Two Micron All Sky Survey (2MASS) (Skrutskie et al. 2006). A left-join cross-match[2] was performed using the pre-computed *Gaia* DR3 cross-match (Marrese et al. 2022), with the `tmass_psc_xsc_best_neighbour` table and a separation ≤ 0.3 arcsec, giving 5333 cross-matched sources.

3. Analysis framework and cluster membership

3.1. *The* COSMIC *pipeline*

The Characterization Of Star clusters using Machine learning Inference and Clustering (COSMIC)~~is an~~[3] is the open-source PYTHON suite ~~that applies unsupervised machine learning and Bayesian estimation to identify open-cluster parameters from large surveys such as *Gaia*~~developed for this work; it chains the membership determination and the Bayesian parameter estimation described below into a single reproducible pipeline, using PyMC[4] ~~for the Bayesian modeling~~(Abril-Pla et al. 2023) for all Bayesian models.

~~For sampling we use the~~ The analysis proceeds in five steps, each performed by a dedicated, publicly available tool. (i) Cluster members are identified as an over-density in the proper-motion plane with the hierarchical density-based clustering algorithm HDBSCAN[5] (Campello et al. 2013; McInnes et al. 2017) (Sect. 3.2). (ii) From the resulting members, the mean parallax and distance, the mean proper motion, and the King structural parameters are inferred with Bayesian models sampled with the No-U-Turn ~~Sampler (NUTS),~~ Sampler (NUTS; Hoffman & Gelman 2014). NUTS is a Hamiltonian Monte Carlo variant that adapts its trajectory

[2] The left join keeps all *Gaia* sources and appends the available 2MASS data for those with a counterpart, preserving the original 15276 sources.

[3] https://github.com/notluquis/COSMIC

[4] https://www.pymc.io

[5] https://hdbscan.readthedocs.io

length by terminating when the path begins to fold back on itself, ~~providing efficient exploration of~~ and it explores high-dimensional posteriors efficiently without the tuning constraints of Metropolis-Hastings or Gibbs sampling ~~. NUTS is used for the parallax, distance, proper-motion, and King-profile models; the isochrone fit instead uses the gradient-free differential-evolution Metropolis (DEMetropolis)ensemble sampler ,~~ (Abril-Pla et al. 2023). (iii) The cluster center is located with a weighted kernel density estimate from SCIKIT-LEARN[6] (Pedregosa et al. 2011). (iv) The age, metallicity, extinction, and distance modulus are obtained by Bayesian isochrone fitting of the member photometry with ASteCA[7] (Perren et al. 2015) (Sect. 6.1); because its grid-interpolated likelihood lacks usable gradients~~.~~, this model is sampled with the gradient-free differential-evolution Metropolis ensemble sampler (DEMetropolis; ter Braak 2006) instead of NUTS. (v) Finally, the pre-main-sequence content of the membership is characterized with the SAGITTA neural network[8] (McBride et al. 2021) (Sect. 6.2). Each step operates on the output of the previous one: the HDBSCAN membership defines the sample on which all subsequent astrometric, structural, photometric, and PMS analyses are performed.

3.1.1. ~~Membership determination~~

All NUTS-based models were run with four chains of 2000 draws after 2000 tuning steps and satisfy the standard convergence criteria (Vehtari et al. 2021): the rank-normalized $\hat{R}$, which compares the between- and within-chain variances and approaches unity when all chains sample the same posterior, is ≤ 1.005 for every parameter (recommended: < 1.01); the bulk and tail effective sample sizes (ESS), i.e. the number of effectively independent draws available for the posterior center and its quantile extremes after accounting for autocorrelation, are above 1200 (recommended: > 400); there are no divergent transitions (numerical failures of the Hamiltonian integrator that can bias the sampled posterior); and the energy Bayesian fraction of missing information (E-BFMI; Betancourt 2016), which measures how efficiently the sampler explores the posterior energy distribution, is above 0.79, well clear of the empirical ~0.3 level below which exploration becomes unreliable (Betancourt 2017). For the DEMetropolis isochrone fit (300 chains of 1000 draws after 1500 tuning steps), to which these per-chain criteria do not directly apply, we verified that the posterior summaries are stable under random half-ensemble bootstraps: the mode scatter is 0.08 dex in log(age) and 0.03 mag in distance modulus, well below the quoted credible widths.

Unless stated otherwise, the values below are derived for the $\tilde{p} \geq 0.6$ reference sample (the membership pseudo-probability $\tilde{p}$ is defined in Sect. 3.2), with the point estimator (mode or median) stated where each fit is described. Table 1 summarizes the parameters derived in the following sections.

3.2. *Membership determination*

To identify potential members of NGC 6383, we employed the Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) algorithm (Campello et al. 2013; McInnes et al. 2017), which identifies clusters of arbitrary shape

Table 1. Summary of derived parameters for NGC 6383, focusing on sources with at least a 60% membership pseudo-probability. Unless otherwise stated, the quoted uncertainties correspond to the 1σ posterior uncertainty or to the 1σ dispersion of the relevant member distribution. For the astrometric quantities, the proper-motion uncertainties are the observed member dispersions, and the parallax uncertainty is the 1σ dispersion of the subsample with fractional parallax errors below 0.1; the smaller errors on the mean are given in the text where relevant. The $Y_{\rm frac}$ uncertainties are the 95% credible-interval bounds of the Beta posterior, not the 1σ default. The parallax-based distance is the mode of the hierarchical distance model applied to the Bailer-Jones et al. (2021) geometric distances of the members (Sect. 4.1); the distance from the distance modulus is obtained by converting the isochrone-fit distance-modulus posterior (Sect. 6.1), and is not independent of the parallax-based value because the distance-modulus prior was centered on it. R_c, k, and b are from the King fit to the 40 arcmin reference sample; R_t is from the 70 arcmin-window refit (Appendix D), the only extraction whose profile reaches a fitted background annulus beyond the cluster; all narrower windows give consistent R_t values. C combines the adopted R_c and R_t posteriors, which come from these two different fits.

Parameter	Value	Unit
Distance (parallax-based)	1.11 ± 0.06	kpc
Distance (from distance modulus)	1.15 ± 0.05	kpc
Distance modulus	10.30 ± 0.09	mag
Age	$3.53^{+1.40}_{-1.00}$	Myr
Star-formation age range (isochrone bracketing)	1.6–6.3	Myr
Metallicity (Z)	0.024 ± 0.008	-
Parallax (ϖ)	0.908 ± 0.046	mas
Reference sample	254	stars
Absorption (A_V)	1.24 ± 0.26	mag
Galactocentric distance $R_{\rm GC}$	7.19 ± 0.06	kpc
Core radius (R_c)	$1.96^{+0.19}_{-0.16}$	arcmin
Background (b)	0.011 ± 0.007	stars arcmin^{-2}
King outer radius (R_t)	54^{+7}_{-11}	arcmin
Center density (k)	$4.92^{+0.43}_{-0.31}$	stars arcmin^{-2}
Hill radius	33.6 ± 2.2	arcmin
Gravitational bound radius	42.5 ± 1.6	arcmin
Cluster center R.A.	263.683 ± 0.112	deg
Cluster center Dec.	-32.584 ± 0.112	deg
Proper motion R.A.	2.542 ± 0.152	mas yr^{-1}
Proper motion Dec.	-1.713 ± 0.138	mas yr^{-1}
Half-mass relaxation time	24.7 ± 7.6	Myr
Radial velocity	-9.4 ± 32.1	km s^{-1}
Concentration parameter (C)	$1.43^{+0.07}_{-0.10}$	-
Structure parameter (Q)	1.03 ± 0.03	-
Mean stellar mass $\langle m \rangle$	1.31 ± 0.10	$M_\odot$
Most massive star	13.56 ± 3.25	$M_\odot$
Young stellar objects (YSOs)	53/193	stars
$Y_{\rm frac}$	$0.275^{+0.067}_{-0.058}$	-
Minimum segregation time	2.38 ± 1.24	Myr

and density without a predetermined cluster number and is robust against noise and outliers (Hunt & Reffert 2021).

After narrowing the data to a final subset of 15276 sources, HDBSCAN was applied only in the two-dimensional proper-motion plane ($\mu_{\alpha*}, \mu_\delta$). Parallax, angular position, and photometry were not used as clustering coordinates; they were used only in the filtering and subsequent characterization steps described below. The clustering used the Euclidean metric in proper-motion space. We also tested the Haversine metric, but it produced nonphysical groupings for this dataset.

To refine our selection of hyperparameters for optimal cluster identification, we tested the grid $m_{\rm cl} = 10, \ldots, 299$, setting `min_cluster_size=min_samples=`$m_{\rm cl}$ in each run, and inspected both the number of recovered sources and the stability of the selected cluster.

The HDBSCAN condensed tree is parameterized by the density level $\lambda = 1/d_{\rm mreach}$, where $d_{\rm mreach}$ is the mutual-reachability distance~~;~~, so that larger λ ~~values correspond to denser~~ corresponds to denser regions of the proper-motion plane. This parameter enters the analysis in three specific, limited

[6] https://scikit-learn.org
[7] https://asteca.github.io
[8] https://github.com/hutchresearch/Sagitta

ways; it is never a fitted physical parameter of the cluster. First, within each sweep run the NGC 6383 candidate branch is identified as the condensed-tree branch that contains the highest density level $\lambda_{\max}$ of the run, that is, the densest proper-motion over-density. Second, λ defines the per-source membership strength $p_{\mathrm{HDBSCAN},i}$ that enters the pseudo-probability defined below. Third, $\lambda_{\max}$ serves as a stability diagnostic of the hyperparameter sweep: the adopted `min_cluster_size` is the value that maximizes the size of the recovered branch, ~~smaller-distance levels in the hierarchy. The `lambda_val` values inspected in the sweep are therefore diagnostics of branch persistence and separation, not fitted physical parameters of the cluster.~~ with $\lambda_{\max}$ acting only as a tie-breaker, and the sweep diagnostic in Appendix B displays only the stable regime with $\lambda_{\max} \geq 8$ and a recovered branch of 200–701 sources. This display cut affects neither the adopted configuration nor the membership.

We adopted `min_cluster_size`=43, `min_samples`=43, and the `leaf` cluster-selection method. These settings maximize the recovered NGC 6383 branch while preserving a high separation from the field population in the condensed tree. The variation in cluster size as a function of the minimum cluster size is included with the HDBSCAN diagnostics in Appendix B.

Because a single HDBSCAN run can overstate the significance of dense fluctuations in noisy data (Hunt & Reffert 2021), we derived a membership pseudo-probability that combines two factors obtained from the same $m_{\mathrm{cl}} = 10, \ldots, 299$ sweep. ~~For each source, we computed~~ The first factor is the recovery frequency f_i, defined as the ~~number of iterations in which the source was assigned to~~ fraction of the ~~selected NGC 6383 branch divided by the total number of iterations, and multiplied it by the native~~ 290 sweep runs in which source i is assigned to a cluster rather than classified as noise; it measures the robustness of the assignment against the choice of m_{cl}. The second factor is the cluster-membership strength $p_{\mathrm{HDBSCAN},i}$ reported by HDBSCAN ~~membership probability $p_{\mathrm{HDBSCAN},i}$. The final value is therefore~~ for the adopted configuration ($m_{\mathrm{cl}} = 43$). HDBSCAN computes it from the condensed tree as $p_{\mathrm{HDBSCAN},i} = \lambda_i/\lambda_{\max}$, where λ_i is the density level at which source i leaves its cluster and $\lambda_{\max}$ is the highest density level attained within that cluster (McInnes et al. 2017). This strength equals 1 for sources that persist to the densest core of the over-density, decreases toward 0 for sources at the cluster boundary, and is 0 for noise. The membership pseudo-probability is the product $\tilde{p}_i = f_i\, p_{\mathrm{HDBSCAN},i}$, so that a high value requires a source to be both stably recovered across the sweep and located close to the density core of the adopted clustering. This quantity is an operational ranking statistic and not a calibrated posterior membership probability. Sources of the selected NGC 6383 branch with $\tilde{p} > 0.5$ were retained as likely cluster candidates. The reference sample used for most cluster parameters adopts $\tilde{p} \geq 0.6$; sources with $0.6 \leq \tilde{p} < 0.8$ are referred to as probable members, while sources with $\tilde{p} \geq 0.8$ are referred to as members. These tiers are a priori round-number conventions rather than tuned values; the stability of the derived parameters against the threshold choice is verified for the mass-segregation analysis (Sect. 7) and for the membership itself through the sweep-marginalized pseudo-probabilities.

Finally, we used the AstroPy Sigma Clipping (Astropy Collaboration et al. 2013, 2018, 2022) utility to apply a 2σ outlier rejection around the mode of the parallax distribution, a conventional rejection level chosen a priori. This step provided the sample of likely cluster candidates from which the $\tilde{p} \geq 0.6$ reference sample used for the cluster characterization was drawn.

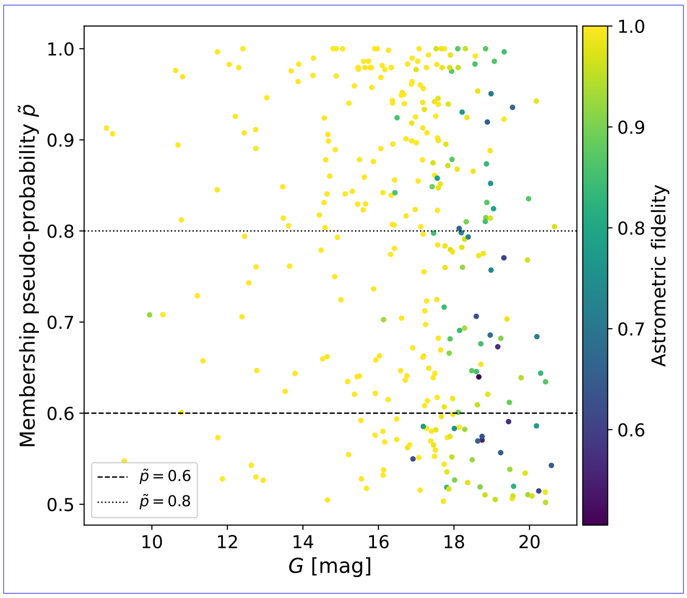

Fig. 1. ~~Radial density profile~~ Membership pseudo-probability $\tilde{p}$ of potential NGC 6383 ~~in logarithmic scale: annular number densities of the reference sample~~ members (~~black points~~$\tilde{p} > 0.5$) ~~with the King model at the posterior median~~ against their G magnitudes; each point is a star, color-coded by astrometric fidelity (~~blue solid line~~viridis color map; color bar at right)~~and its 16–84% posterior band (blue shading)~~. The ~~black~~ horizontal dashed and dotted ~~vertical~~ lines, at $\tilde{p} = 0.6$ and $\tilde{p} = 0.8$ respectively, mark the ~~core radius~~ thresholds used to distinguish between probable members (~~$R_c = 1.95$ arcmin~~$0.6 \leq \tilde{p} < 0.8$) and ~~the fitted King outer radius~~ members (~~$R_t = 40.4$ arcmin~~$\tilde{p} \geq 0.8$) ~~, and~~ of the ~~gray dash-dotted horizontal line marks the fitted background level b~~cluster. ~~The posterior band is narrow near~~ As shown in the ~~core~~figure, ~~where the profile is set by the well-constrained central density~~stars with fainter magnitudes tend to have lower astrometric fidelity. This becomes more evident at $G > 18$ mag.

The 40 arcmin run defines the production catalog used throughout this paper. To test whether the membership recovery was limited by this original search field, we repeated the complete preprocessing, HDBSCAN sweep, pseudo-probability calculation, and parallax clipping using input radii of 50, 60, and 70 arcmin. ~~These wider-field runs are diagnostic checks only; the~~ The adopted cluster parameters remain tied to the 40 arcmin reference catalog~~unless explicitly stated.The~~, except the King outer radius, which is measured on the widest extraction (Sect. 5); the setup and source counts of the robustness runs are reported in Sect.~~3.3.~~ 3.3.

3.3. Membership of NGC 6383

Applying the HDBSCAN algorithm in the proper-motion plane returned several clusters. Given HDBSCAN's tendency to report dense random fluctuations as clusters (Sect. 3.2), we selected the branch that is both stable in the condensed cluster tree and centered on the known proper-motion overdensity of NGC 6383, which stands out as a compact, isolated clump in the proper-motion plane while overlapping the field on the sky (Fig. B.3, Appendix B). This cluster initially contained 701 sources, and after applying the 2σ parallax clipping described in Sect. 3.2, 321 sources were retained.

Regarding membership pseudo-probabilities (Fig. 1), NGC 6383 has 321, 254, 202, and 161 candidate members with

$\tilde{p}$ above 50, 60, 70, and 80%, respectively. Correspondingly, 288, 236, 191, and 153 of these candidates are brighter than $G = 19$ mag. Table 2 shows a five-row excerpt of the likely-member catalog; the complete 321-source catalog is available at the CDS (see Data availability), with the full column set listed in the table note and the CDS ReadMe.

The cone-search robustness experiment (Appendix D, Table D.1) shows how this selection behaves when the input field is widened; the corresponding astrometric and King-structural parameters recovered at each radius are compiled in Table D.2. The NGC 6383-like proper-motion branch is recovered in every run, but the recovered reference sample is not invariant with radius. In the 40 and ~~broadens markedly~~ 50 arcmin runs the generic sweep-selected branch coincides with the NGC-like branch. In the 60 and 70 arcmin runs, however, the generic branch selected by the sweep differs from the branch closest to the NGC 6383 proper-motion overdensity, showing that the automated branch selection becomes field-window dependent in larger cones.

At the catalog level the solution is stable: the 50 arcmin run, in which the automated sweep selects the same branch as the NGC-like criterion, returns mean parallax, mean proper motions, and refitted King parameters consistent with the adopted values (Table D.2). At 60 and 70 arcmin, where the sweep-selected branch diverges from the NGC-like branch and the $\tilde{p} \geq 0.6$ sample inflates to 443 and 628 sources, the solution degrades coherently: the proper-motion dispersions grow and the means drift (Table D.2), and the refitted profile reflects the added contamination through a smaller R_c, while the outer radius, now background-constrained, reaches 54^{+7}_{-11} arcmin (Sect. 5, Appendix D).

At the source level, the runs agree well: the wider fields recover 203 (80%), 242 (95%), and 252 (99%) of the adopted 254 at ~~large radii~~50, ~~where it is governed by the weakly constrained outer radius R_t~~ 60, and 70 arcmin, and 192 sources (76%) appear in all four runs. The 50 arcmin run adds only 33 sources, all but one beyond the 40 arcmin footprint. The 60 and ~~background level b.~~70 arcmin runs, by contrast, add 201 and 376 sources, of which 83 and 114 lie *inside* the 40 arcmin footprint, the signature of a lowered effective selection threshold rather than of newly resolved outer structure. This behavior is consistent with increasing field contamination, although distinguishing contaminants from genuinely extended populations would require additional photometric, dynamical, or three-dimensional diagnostics (Sect. 5). We therefore use the 40 arcmin catalog as the reference membership sample, and the wider extractions to constrain the outer structure (Sect. 5), while treating the outer membership and any quantity tied to the cluster boundary as search-window sensitive.

3.3.1. ~~Parallax and distance estimation~~

4. Astrometric parameters

4.1. Parallax and distance

Distance estimation from parallax is challenging due to measurement errors, making the conversion from parallax to distance via the simple inverse method unreliable (Bailer-Jones 2015; Bailer-Jones et al. 2021). Choosing an appropriate prior distribution is crucial. To address this challenge, we obtained photo-geometric and geometric distances from Bailer-Jones et al. (2021), who introduced a Bayesian approach incorporating a prior distribution tailored to different regions of the sky, improving distance estimates from *Gaia* EDR3 data. However, we opted not to use photogeometric distances due to the presence of PMS stars and gas in the cluster, which could distort the magnitudes and colors.

When analyzing the parallax measurements of stars within an open cluster, and assuming a simplified one-dimensional perspective, we assume that the parallax values follow a normal distribution, $\varpi_i \sim \mathcal{N}(\mu_\varpi, \sigma_\varpi)$, ~~with $\mu_\varpi \sim \mathcal{U}(0.5\bar{\varpi}, 1.5\bar{\varpi})$ and σ_ϖ given a half-normal priorwith the frequentist dispersion as scale .~~where $\mathcal{N}(\mu, \sigma)$ denotes a normal distribution with mean μ and standard deviation σ. The mean was given a uniform prior, $\mu_\varpi \sim \mathcal{U}(0.5\,\bar{\varpi}, 1.5\,\bar{\varpi})$, where $\mathcal{U}(a, b)$ denotes the uniform distribution on the interval $[a, b]$ and $\bar{\varpi}$ is the arithmetic mean of the observed parallaxes of the sources entering the fit (those with fractional parallax error below 0.1; see below). The dispersion was given a half-normal prior, $\sigma_\varpi \sim \mathcal{HN}(s_\varpi)$, where $\mathcal{HN}(s)$ denotes the half-normal distribution with scale s and s_ϖ is the standard deviation of the same observed parallaxes. We use the $\mathcal{N}$, $\mathcal{U}$, and $\mathcal{HN}$ notation for all priors in the remainder of the paper. Additionally, we assume a gamma distribution for the distances, parameterized by its mean μ_r and width σ_r, the latter with a half-normal prior~~scaled to the frequentist dispersion~~ , $\sigma_r \sim \mathcal{HN}(s_r)$, with s_r the standard deviation of the geometric distances ~~.~~introduced above.

~~To address this challenge, we obtained photo-geometric and geometric distances from , who introduced a Bayesian approach incorporating a prior distribution tailored to different regions of the sky, improving distance estimates from *Gaia* EDR3 data. However, we opted not to use photogeometric distances due to the presence of PMS stars and gas in the cluster, which could distort the magnitudes and colors.~~

In our hierarchical model, we calculate the mean (μ_{prior}) between the frequentist average distance derived from parallaxes and the Bailer-Jones et al. (2021) geometric distances, using $\mathcal{U}(0.5\mu_{\text{prior}}, 1.5\mu_{\text{prior}})$ as the prior for the cluster's mean distance. Our analysis is limited to sources with a fractional parallax error below 0.1, i.e. $\delta\varpi/\varpi < 0.1$ where $\delta\varpi$ is the *Gaia* parallax uncertainty, to ensure accuracy.

4.1.1. ~~Proper motions~~

The mean parallax of the subsample used for the distance estimate is 0.908 mas, with a 1σ dispersion of 0.046 mas and a standard error on the mean of 0.004 mas (Fig. 2). The mode sampled distance is 1.11 ± 0.06 kpc. These values slightly deviate from the results obtained in earlier studies (Trumpler 1930; Sanford 1949; Fitzgerald et al. 1978; The et al. 1985; Paunzen et al. 2007) but agree with Becker & Fenkart (1971) and with later determinations (Kharchenko et al. 2005; Piskunov et al. 2008; Jaehnig et al. 2021; He et al. 2022; Pulgar-Escobar & Henríquez-Salgado 2024; Hunt & Reffert 2024).

4.2. Proper motions and radial velocities

We used a two-dimensional Gaussian to model the distribution of the proper motions of the cluster members. A normal prior distribution was assigned to the mean proper motions $[\overline{\mu_\alpha}, \overline{\mu_\delta}]$, based on frequentist means and standard deviations. For the standard deviations, we used half-normal priors with a width equal to the frequentist standard deviation of the proper motions, and the correlation coefficient between the proper motions was uniformly distributed over the interval $[-1, 1]$. Additionally, we es-

Table 2. Excerpt of the likely-member catalog for NGC 6383. The complete machine-readable table is available at the CDS. Only selected columns and five rows are printed here for guidance regarding the table form and content.

GaiaDR3	RAdeg (deg)	DEdeg (deg)	Gmag (mag)	pHDBSCAN	pFreq	pMember	Ref
4053793547964747904	264.16939344	-33.03325935	17.9052	0.7959	0.9793	0.7795	1
4054500465190725632	263.10710918	-32.96962071	20.1954	0.6985	0.9793	0.6840	1
4054501599068361216	263.08486813	-32.91805440	15.1189	0.8585	0.9793	0.8407	1
4054514449600386304	263.64147648	-33.22555846	16.3805	0.9110	0.9828	0.8953	1
4054518023012382592	264.01601176	-33.15455344	14.6423	0.6760	0.9793	0.6620	1

Notes. The complete CDS table contains the following columns: GaiaDR3, RAdeg, DEdeg, Plx, e_Plx, pmRA, e_pmRA, pmDE, e_pmDE, Gmag, BPmag, RPmag, Jmag, Hmag, Ksmag, Fidelity, pHDBSCAN, pFreq, pMember, Ref, PMSProb, AvSag, and logAgeSag. Here pMember is the composite membership pseudo-probability $\tilde{p} = p_{\rm HDBSCAN} f_i$. The column Ref is equal to one for sources belonging to the $\tilde{p} \geq 0.6$ reference sample used for the main cluster parameters, and zero otherwise.

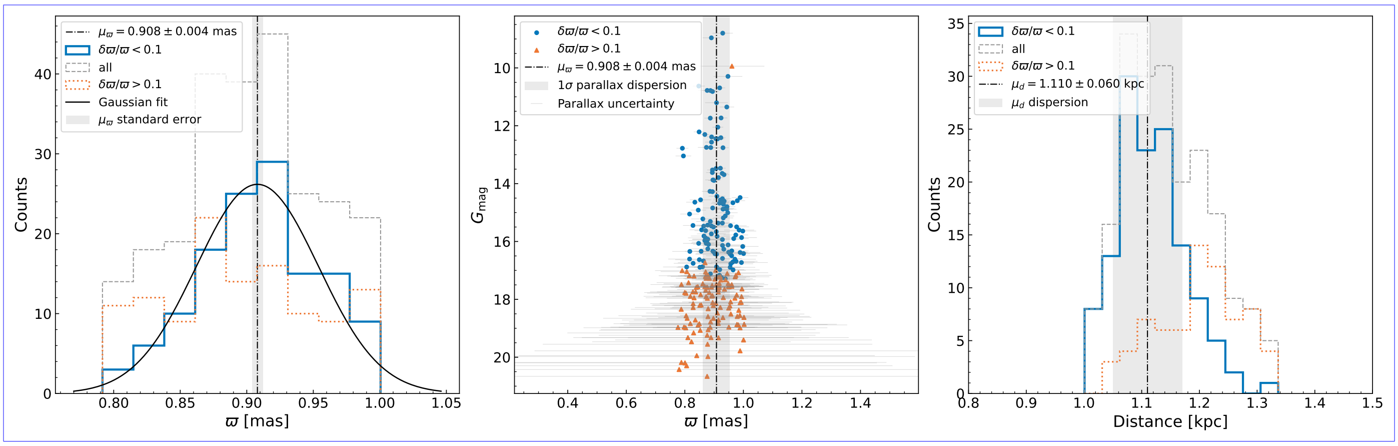

Fig. 2. *Left panel:* Parallax distribution of the 254-source reference sample, split by fractional parallax error: sources with $\delta\varpi/\varpi < 0.1$ (blue solid histogram; used for the estimate), all sources (gray dashed), and $\delta\varpi/\varpi > 0.1$ (orange dotted; not used). The black curve is a Gaussian fit; the black dash-dotted line and the gray vertical band mark the mean parallax, $\mu_\varpi = 0.908$ mas, and its standard error. *Middle panel:* G magnitude versus parallax with the same split (blue circles < 0.1, orange triangles > 0.1); gray bars show the parallax uncertainties, especially pronounced at fainter magnitudes; the mean-parallax line is repeated, with a gray band showing the 1σ parallax dispersion. *Right panel:* Bailer-Jones geometric distances (Bailer-Jones et al. 2021) with the same split; the black dash-dotted line marks the adopted distance ($\mu_d = 1.11$ kpc, the mode of the sampled distances), with a gray band showing its ± 0.06 kpc dispersion. For convenience, only the portion of the histogram corresponding to distances within 0.8 to 1.2 times the minimum and maximum used distances is displayed.

timated the mean total proper motion of the cluster, defined as the quadrature sum of the proper-motion components.

~~All NUTS-based models were run with four chains of 2000 draws after 2000 tuning steps and satisfy the standard convergence criteria : rank-normalized $\hat{R} \leq 1.005$ for every parameter, bulk and tail effective sample sizes above 1200, no divergent transitions, and energy-BFMI above 0.79. For the DEMetropolis isochrone fit (300 chains of 1000 draws after 1500 tuning steps)~~The mean proper-motion values are 2.542 mas yr^{-1} in R.A. and -1.713 mas yr^{-1} in Dec., with member dispersions of 0.152 and 0.138 mas yr^{-1}, respectively, which align with the acceptable values from Perren et al. (2023) (Fig. 3). The corresponding standard errors on the means are 0.0096 and 0.0086 mas yr^{-1}, but these smaller values are not used as the physical width of the cluster when comparing individual stars. The posterior of the total proper motion, derived as the quadrature sum of the two components in each posterior draw, has a median of 3.065 ± 0.009 mas yr^{-1}, corresponding to a projected (tangential) velocity of 16.1 ± 0.9 km s^{-1} at the adopted distance.

Gaia DR3 radial velocities are available for only 11.4% of the reference sample (29 of 254 sources). Because sources with large radial-velocity amplitudes, likely unresolved binaries, can bias the cluster's radial-velocity estimate, we restricted it to the 16 clustered sources with $\tilde{p} \geq 0.6$ and a binary probability (an ASteCA product; Sect. 7) lower than 0.6. The resulting median, mean, and standard deviation are -1.87, -9.40, and 32.1 km s^{-1}, respectively. The substantial standard deviation primarily stems from the limited availability of radial velocity data, and even with the binary cut the small sample makes this only an indicative measure of the cluster's internal dynamics. Figure C.6 illustrates all the probable members and members with available radial velocities, their magnitudes, and the amplitude of their radial velocities.

4.3. Cluster center

To determine the cluster's center, we used a weighted Kernel Density Estimation (KDE) from the Python library ~~SCI-KIT LEARN~~SCIKIT-LEARN (Pedregosa et al. 2011), assigning weights inversely proportional to the distance from the mean proper motion. Optimal KDE parameters were determined through Grid Search Cross-Validation within the package, exploring a range of bandwidths ~~—~~(from the mean positional error to the cone-search radius~~—~~) and all kernel types; the selected bandwidth

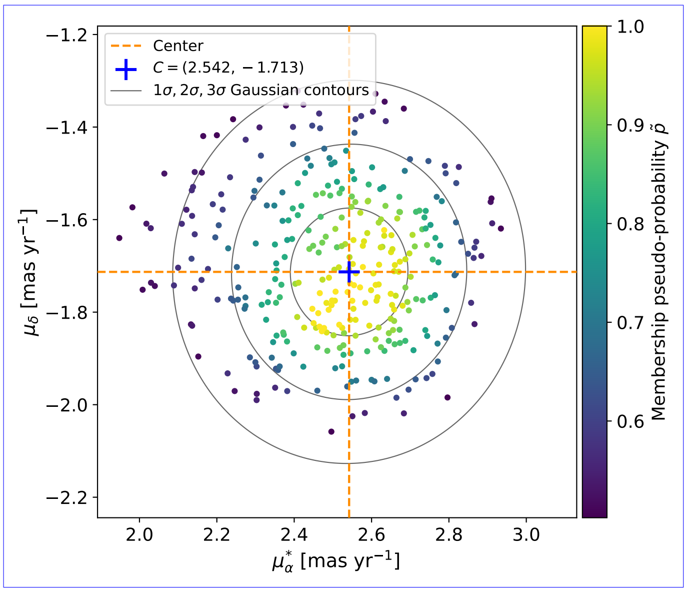

Fig. 3. ~~CMD~~ Proper motions of ~~NGC 6383 ($G_{\rm mag}$ versus $G_{\rm BP}-G_{\rm RP}$) with~~ the ~~Sagitta-based PMS classification. Orange filled symbols are PMS~~ likely cluster candidates (~~Sagitta PMS probability ≥ 0.6~~$\tilde{p} > 0.5$)~~, blue filled~~ of NGC 6383; the symbols are ~~non-PMS sources, and open black symbols are sources without 2MASS photometry, which we exclude from~~ color-coded according to the ~~Sagitta-based classification; star markers denote members~~ membership pseudo-probability $\tilde{p}$ of the stars (~~$\tilde{p} \geq 0.8$~~viridis color map)~~and diamonds probable members ($0.6 \leq \tilde{p} < 0.8$)~~. The ~~black~~ gray solid ~~, dashed, and dotted curves show~~ ellipses are the ~~synthetic single-star sequences at the mode~~1σ, ~~mean~~2σ, and ~~median~~ 3σ iso-probability contours of the ~~isochrone-fit posterior, and~~ two-dimensional Gaussian proper-motion model fitted to the ~~light-gray curves are 150 single-star sequences drawn from~~ reference sample. The center of the ~~isochrone-fit posterior~~proper-motion distribution is marked with a blue cross, ~~visualizing~~ indicating the ~~fit uncertainty~~inferred mean proper-motion values $(\mu_{\alpha*}, \mu_\delta) = (2.542, -1.713)$ mas yr^{-1}, with corresponding orange dashed lines highlighting these central coordinates.

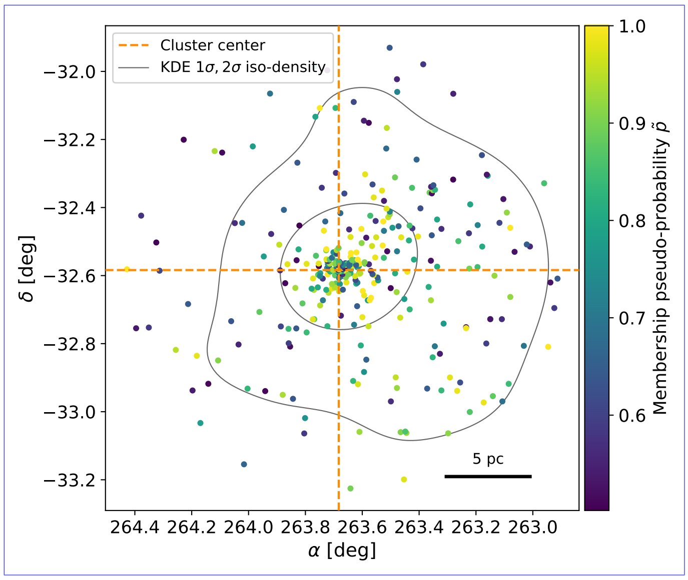

Fig. 4. Spatial distribution of the likely cluster candidates ($\tilde{p} > 0.5$) of NGC 6383 in right ascension (α) and declination (δ), color-coded according to their membership pseudo-probability $\tilde{p}$ (viridis color map). The gray solid contours are the 1σ and 2σ iso-density contours (enclosing 39% and 86%) of a kernel density estimate of the candidate positions, revealing the density gradient of the points. The central coordinates of the cluster are marked by dark-orange dashed lines at $\alpha = 263.683°$ and $\delta = -32.584°$, which approximately coincide with the density peak within the plotted area. A scale bar of 5 parsecs is included at the bottom right to provide a reference for the spatial scale.

($\approx$ 6.7 arcmin) dominates the quoted center uncertainty. The location of maximum density was taken as the cluster's center.

4.3.1. ~~Radial density profile~~

The central position of the cluster in equatorial coordinates is $263.683 \pm 0.112°$ in R.A. and $-32.584 \pm 0.112°$ in Dec. (Fig. 4). These errors are calculated from the quadratic sum of the mean errors in R.A. and Dec. and the optimal KDE bandwidth. This center is consistent, within the quoted uncertainty, with the projected coordinates of HD 159176; the positional coincidence is therefore not sufficient by itself to establish membership.

5. Cluster structure

To determine the structural parameters of the cluster, we utilized the King (1962) density profile. The numerical density ρ ~~,~~was computed by dividing the cluster area into concentric annuli, each containing an equal number of stars. The number of annuli, K, adheres to the equiprobable bin rule ($K = 2n^{2/5}$), where n is the star count within the cluster. The annular densities were fit with a Gaussian likelihood whose scatter is a half-normal nuisance parameter, yielding the posterior distribution of the structural parameters.

Priors for the model parameters were set as follows: $b \sim \mathcal{U}(0, 2\rho_{\rm min})$, $k \sim \mathcal{U}(b, 2\rho_{\rm max})$, $R_c \sim \mathcal{U}(0, 0.8\,T_{\rm max})$, and $R_t \sim \mathcal{U}(R_c, 1.5\,T_{\rm max})$, where b is the background density, k the central normalization, $\rho_{\rm min}$ and $\rho_{\rm max}$ the observed surface-density extrema, and $T_{\rm max}$ the highest value between the Hill radius and the gravitational bound radius~~. This ensures that the chosen parameter space for~~ , both defined below. These coefficients are not tuned values but weakly informative bounds anchored to the largest physically meaningful scale of the system. The upper limit on R_c lies more than an order of magnitude above any realistic open-cluster core radius. The upper limit on R_t allows the fitted outer radius to exceed the nominal gravitational bound radius by up to 50%, generous headroom for the uncertainties in the cluster mass and Galactic potential that enter $T_{\rm max}$.

The posterior of R_c is confined well inside its support (97.5th percentile at $\simeq 2.5$ arcmin, 7% of the prior upper bound), so its bound is immaterial. Refitting the 40 arcmin profile with alternative coefficients (e.g., $0.9\,T_{\rm max}$ and $1.2\,T_{\rm max}$) changes R_c, k, b, and C by less than 0.5σ; only the weakly constrained outer radius responds (median $R_t \simeq 33$ arcmin versus 40^{+16}_{-17} arcmin, consistent within 1σ). The outer radius is the one parameter whose posterior is sensitive to the prior support in all windows: the upper bound $1.5\,T_{\rm max} = 63.7$ arcmin truncates the R_t ~~is within astrophysical valid limits.~~ posterior of every fit. All quoted R_t intervals are therefore conditional on this physically motivated prior (Appendix D), reinforcing its interpretation as a model-dependent outer scale.

The Hill radius ~~($R_{\rm hill}$)~~($R_{\rm Hill}$), which defines the region around a star cluster where its gravitational influence dominates over that of the Galaxy, can be calculated following Carrera et al.

(2019):

$$R_{\cancel{\rm hill}\,{\rm Hill}} = R_{\rm gc} \times \left(\frac{m_c}{3M_{\rm gc}}\right)^{1/3} \tag{1}$$

where ~~$R_{\rm gc} = 7.19 \pm 0.07$ kpc denotes~~ $R_{\rm gc} = \left(R_\odot^2 + d^2 - 2R_\odot d \cos l \cos b\right)^{1/2} = 7.19 \pm 0.06$ kpc is the galactocentric distance ~~,~~ of the cluster, computed from its Galactic coordinates (l, b), the parallax-based distance $d = 1.11 \pm 0.06$ kpc, and a solar galactocentric radius $R_\odot = 8.3$ kpc (de Grijs & Bono 2016), with the uncertainty propagated from the distance and cluster-center uncertainties, m_c the cluster mass from Hunt & Reffert (2024), with a value of 902 ± 92 $M_\odot$, and ~~$M_{\rm gc} = (1.43 \pm 0.02) \times 10^{11}$ M$_\odot$~~ $M_{\rm gc}$ the Galactic mass enclosed within the cluster's orbit, ~~calculated as~~ estimated from the circular velocity of the McMillan (2017) Galactic potential at $R_{\rm gc}$, $V_c(7.19\,{\rm kpc}) \simeq 229$ km s^{-1}, as

$$M_{\rm gc} = \cancel{2}\frac{V_c^2(R_{\rm gc})\, R_{\rm gc}}{G} = (8.8 \pm 0.5) \times 10^{\cancel{8}10}\, M_\odot \cancel{\left(\frac{R_{\rm gc}}{30\,{\rm pc}}\right)^{1.20}}, \tag{2}$$

Moreover, the gravitationally bound radius was estimated using the method described by Pinfield et al. (1998), which considers the cluster mass and Oort constants, using the formula

$$r_{\rm bound} = \left(\frac{G m_{\rm c}}{2(A - B)^2}\right)^{1/3}, \tag{3}$$

where G is the gravitational constant, and A and B are the Oort constants, with values from Bovy (2017).

The King fit to the 40 arcmin reference sample yields a core radius $R_c = 1.96^{+0.19}_{-0.16}$ arcmin ($0.63^{+0.06}_{-0.05}$ pc), with $k = 4.92^{+0.43}_{-0.31}$ and background $b = 0.011 \pm 0.007$ stars per square arcminute (Fig. 5). The outer radius, however, is not usefully constrained by this window: the fit returns $R_t = 40^{+16}_{-17}$ arcmin with half of the posterior mass beyond the extraction radius itself, because the field contains almost no background annulus beyond the cluster.

We therefore measure the outer structure on the wider extractions of Appendix D, refitted with identical priors. The fitted R_t is mutually consistent across all windows (Table D.2), and the 70 arcmin window is the only one whose radial profile reaches a fitted background annulus beyond the cluster (Fig. D.1). We adopt $R_t = 54^{+7}_{-11}$ arcmin ($17.4^{+2.3}_{-3.6}$ pc) as the King outer radius. This interval remains conditional on the $R_t < 1.5\,T_{\rm max} = 63.7$ arcmin prior: relaxing the bound to the extraction edge ($R_t < 70$ arcmin) shifts the posterior to 58^{+9}_{-13} arcmin, consistent within 1σ, with the upper tail still limited by the prior support rather than turned over by the data. The wider windows also admit additional field contamination in the membership (Sect. 3.3), which lowers the fitted R_c to $\simeq 1.4$ arcmin at 60–70 arcmin; the selection-stable 50 arcmin run returns $R_c = 1.90^{+0.15}_{-0.13}$ arcmin, confirming the adopted core radius. Combining the adopted R_c and R_t posteriors gives a concentration parameter $C = \log_{10}(R_t/R_c) = 1.43^{+0.07}_{-0.10}$ (Peterson & King 1975). Because the added outer sources cannot be separated from field contamination without additional diagnostics (Risbud et al. 2025; Xu et al. 2025), R_t remains a model-dependent scale rather than a sharp physical boundary. The half-light radius is determined to be 6.02 ± 1.07 arcmin (1.94 ± 0.36 pc, bootstrap uncertainty) and the half-mass radius is 6.26 ± 1.23 arcmin (2.02 ± 0.41 pc).

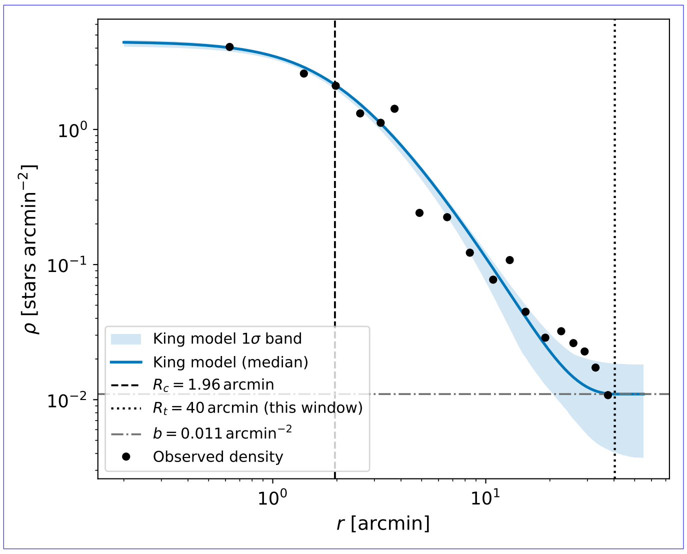

Fig. 5. Radial density profile of NGC 6383 in logarithmic scale: annular number densities of the reference sample (black points) with the King model at the posterior median (blue solid line) and its 16–84% posterior band (blue shading). The black dashed and dotted vertical lines mark the core radius ($R_c = 1.96$ arcmin) and the outer radius preferred by this 40 arcmin-window fit ($R_t = 40$ arcmin), and the gray dash-dotted horizontal line marks the fitted background level b. The posterior band is narrow near the core, where the profile is set by the well-constrained central density, and broadens markedly at large radii, where this window contains almost no background annulus; the adopted outer radius, $R_t = 54^{+7}_{-11}$ arcmin, is measured on the 70 arcmin extraction (Appendix D, Fig. D.1).

Measuring the outer structure on a field substantially larger than the cluster itself follows established *Gaia* DR3 practice (Angelo et al. 2025); even so, the wider-field tests (Appendix D, Table D.1) show that the number of outer candidates changes with the input field, so we do not classify the larger-radius additions as confirmed halo members or contaminants. We identify 18 sources beyond the Hill radius (33.6 arcmin) with cluster-like proper motions (pseudo-probabilities 0.64–1.0, mean 0.81; Fig. C.3); all lie within the 40 arcmin extraction, and hence well inside the adopted R_t. They also lie inside the gravitational bound radius (42.5 arcmin); we nevertheless flag them because the Hill radius, which accounts for the Galactic tidal field at the cluster's orbit, is the more conservative boundary of guaranteed boundness, so their membership in the bound system is not certain.

6. Age, extinction, and stellar content

6.1. *Isochrone fitting with ASteCA*

Automated Stellar Cluster Analysis (ASteCA) (Perren et al. 2015), is an open-source Python tool designed to automate standard tests for determining basic parameters of stellar clusters. For accurate estimation of a cluster's metallicity, age, and extinction values, we employed ASteCA's isochrone fitting process. We utilized the MIST isochrones (Dotter 2016; Choi et al. 2016) along with the photometric systems of *Gaia* EDR3 and 2MASS. The Initial Mass Function followed Chabrier et al. (2014). Synthetic binaries were generated with the mass-dependent binary fraction adopted in ASteCA, $b(m) = \alpha + \beta/(1 + 1.4/m)$ with ~~$\alpha = 0.090$ and $\beta = 0.940$ (rising from $\simeq 0.09$ at low masses toward unity for the most~~

~~massive stars), and~~ $\alpha = 0.090$ and $\beta = 0.940$ held fixed; this parametrization is ASteCA's analytic fit to the observed multiplicity fraction as a function of primary mass compiled by Offner et al. (2023), and with these coefficients it reproduces the observed binarity across spectral types, rising from $\simeq 0.26$ for M dwarfs through $\simeq 0.5$ for solar-type stars to $\gtrsim 0.9$ for O-type stars (Duchêne & Kraus 2013; Moe & Di Stefano 2017). Secondary masses were drawn from the mass-ratio distribution $f(q) \propto q^{\gamma}$ with the mass-dependent exponents of Duchêne & Kraus (2013); differential reddening was fixed to zero.

The MIST ages inherit the physical assumptions of the underlying models: we used the v1.2 grids, which adopt solar-scaled abundances on the Asplund et al. (2009) scale ($Z_{\odot} \simeq 0.0142$), a single solar-calibrated mixing-length parameter ($\alpha_{\rm MLT} = 1.82$), exponentially decaying convective overshooting ($f_{\rm ov,core} = 0.016$, $f_{\rm ov,env} = 0.0174$), and an initial rotation of $v/v_{\rm crit} = 0.4$ imposed at the zero-age main sequence, so the PMS tracks that dominate our fit are effectively non-rotating. The models include neither magnetic inhibition of convection, nor starspots, nor an accretion history; magnetic and spotted PMS models yield systematically older ages for young clusters, by a factor of ~ 2 or more (Feiden 2016; Somers & Pinsonneault 2015), with the SPOTS grids showing the same systematic (Somers et al. 2020), so the absolute PMS age scale is tied to these assumptions.

The priors for metallicity Z, logarithmic age, and absorption A_V were defined as uniform distributions ranging from [0.001, 0.045], [6.00, 7.00], and [0.5, 2], respectively, aligned with the findings of Pulgar-Escobar & Henríquez-Salgado (2024). The distance modulus was modeled as $\mathcal{N}(10.3, 0.2)$, with the mean value corresponding to the expected distance modulus obtained from the parallax. We used only the MIST PMS evolutionary models for the present fit; therefore, the quoted age uncertainty reflects the posterior uncertainty within this model family and does not include the systematic differences that can arise from alternative PMS tracks. We adopted the mode of the posterior distribution for our analysis and sampled the model with the gradient-free DEMetropolis ensemble sampler ~~from PyMC, whose gradient-free proposals are suited to the grid-interpolated isochrone likelihood,~~ (Sect. 3.1) instead of the Approximate Bayesian Computation (ABC) suggested by ASteCA. The nuisance parameter σ shown in the posterior ~~diagnostic plot~~ corner plot (Fig. B.4) is the likelihood-scatter term ~~used in the PyMC diagnostic implementation that generated this figure~~of the fit; it is not a physical velocity, spatial, or age dispersion of the cluster.

To determine the cluster's age, extinction, and metallicity, we fit MIST isochrones to the CMD of NGC 6383 (Fig. 6, which shows the PMS stars and the mode, mean, and median isochrones). The mode indicates a logarithmic age in years of 6.55 ± 0.15, a distance modulus of ~~10.30 ± 0.26 mag~~ 10.30 ± 0.09 mag corresponding to a distance of ~~$1.15^{+0.15}_{-0.13}$ kpc~~1.15 ± 0.05 kpc, and a metallicity Z of 0.024 ± 0.008, slightly above the MIST solar value ($Z_{\odot} \simeq 0.0142$); we note that the absolute metallicity is only loosely constrained by the isochrone fit because of the age–metallicity–extinction degeneracy, and the value lies within the adopted prior. An estimated extinction correction of $A_V = 1.24 \pm 0.26$ was applied. This CMD-derived distance is consistent with the parallax-based value of 1.11 ± 0.06 kpc (Table 1)~~– as expected ,~~, as expected given that the distance-modulus prior was centered on it~~–~~, and we adopt the parallax value for all structural-radius conversions.

The CMD, particularly for sources brighter than $G = 12.0$ mag, displays a well-defined main sequence. The presence of PMS stars and lower-mass main-sequence stars broadens the lower part of the CMD, complicating the fit of a single isochrone. Figure 7 presents various isochrones for *Gaia* and 2MASS data, considering ages between 6.20 and 7.00 in logarithmic scale and initial metallicities Z of 0.024. The isochrones that best bracket the PMS locus correspond approximately to log(age) = 6.20–6.80, equivalent to about 1.6–6.3 Myr, although older isochrones are plotted to show the sensitivity of the fit. Unresolved binaries, differential reddening (fixed to zero in the fit), and accretion variability also broaden the PMS locus (Da Rio et al. 2010), so this bracketing range should be read as an upper limit on any true age spread.

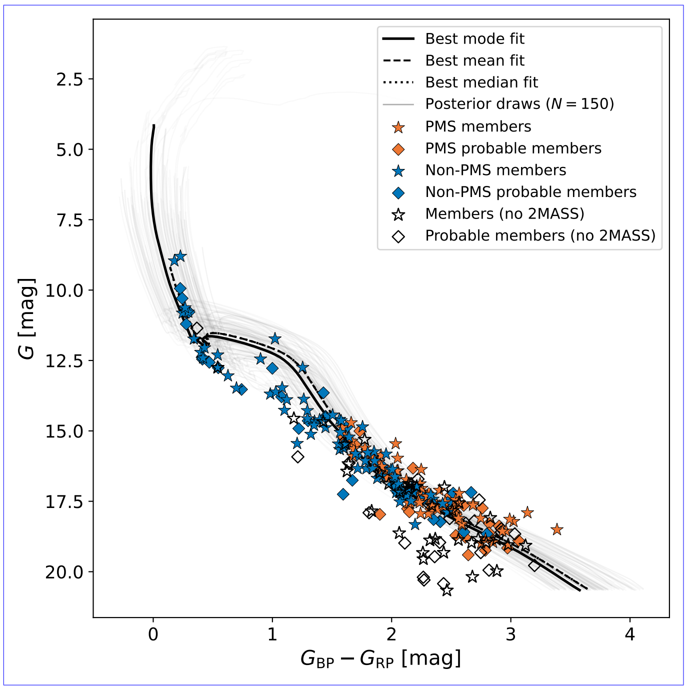

Fig. 6. CMD of NGC 6383 ($G_{\rm mag}$ versus $G_{\rm BP} - G_{\rm RP}$) with the Sagitta-based PMS classification. Orange filled symbols are PMS candidates (Sagitta PMS probability $\gtrsim 0.6$), blue filled symbols are non-PMS sources, and open black symbols are sources without 2MASS photometry, which we exclude from the Sagitta-based classification; star markers denote members ($\tilde{p} \geq 0.8$) and diamonds probable members ($0.6 \leq \tilde{p} < 0.8$). The black solid, dashed, and dotted curves show the synthetic single-star sequences at the mode, mean, and median of the isochrone-fit posterior, and the light-gray curves are 150 single-star sequences drawn from the isochrone-fit posterior, visualizing the fit uncertainty.

Figure 7 also includes color-color diagrams ($G_{\rm BP} - G_{\rm RP}$) showing a clearer segregation of PMS stars, which tend to be at redder colors, as expected. ~~Additionally, Fig. C.4 shows the returned Sagittaparameters, which include PMS probability, age, and A_V.~~The posterior distribution of the inferred parameters can be seen as a plot pair in Fig. B.4.

6.2. *PMS content and YSO fraction*

Sagitta is a neural network designed to classify stars as PMS and estimate their ages. It is trained with data from *Gaia* DR2 and the 2MASS. It incorporates three convolutional neural networks each dedicated to determining stellar extinction (A_V), PMS probability, and estimating stellar ages. Sagitta's age estimation is applicable for young stars up to ~ 80 Myr and relies on inputs

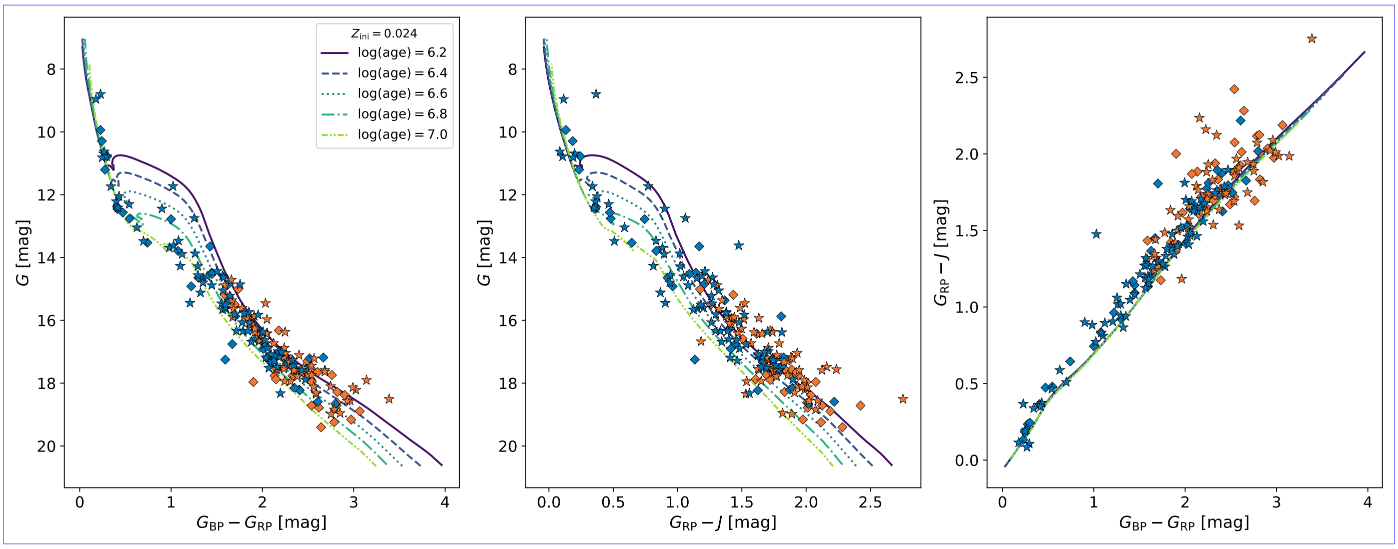

Fig. 7. *Left panel:* CMD of NGC 6383 showing $G_{\rm mag}$ against $G_{\rm BP} - G_{\rm RP}$ with isochrones for log(age yr^{-1}) ranging from 6.20 to 7.00. *Middle panel:* CMD using $G_{\rm mag}$ versus $G_{\rm RP} - J$. *Right panel:* Color-color diagram, $G_{\rm RP} - J$ against $G_{\rm BP} - G_{\rm RP}$, with the equivalent color-color isochrones. The symbol notation follows that of Fig. 6, representing different member classifications and evolutionary states within NGC 6383. The MIST isochrones for log(age) = 6.2–7.0 (at $Z_{\rm ini} = 0.024$) are drawn with sequentially ordered colors and distinct line styles, identified in the left-panel legend, highlighting the evolutionary progression of members within NGC 6383.

including *Gaia* parallaxes, and average line-of-sight extinction (McBride et al. 2021).

The tool's efficacy stems from its training on a well-curated dataset of PMS stars within moving groups (Kounkel et al. 2020), which is supplemented by various literature sources with previously measured ages. Sagitta uniquely offers stability in age predictions across G, K, and M-type stars within the same population, surpassing traditional isochrone techniques in consistency (Kounkel et al. 2023). Although the network was trained on *Gaia* DR2 photometry, we apply it to the corresponding *Gaia* DR3 bands.

Identifying PMS stars and estimating their ages traces the cluster's star-formation history. Sagitta is most reliable for PMS stars, as the photometry of higher-mass stars that have reached the main sequence is not indicative of their age, limiting the model's applicability to stars that remain in the PMS stage.

Applying Sagitta to the membership yields 116 of the 254 reference-sample sources (133 of the 321 likely candidates) with a PMS probability of at least 0.6, the threshold used for the PMS class throughout this paper. The distributions of the per-star Sagitta PMS probabilities, ages, and extinctions are shown in Fig. C.4. These histograms are individual-star Sagitta estimates and should not be read as the same posterior summarized in Table 1~~. The~~: the broader Sagitta distributions, including the high-extinction tail and older age tail, reflect star-to-star extinction, photometric scatter, unresolved binaries, and the fact that Sagitta is applied to individual PMS candidates, whereas the ASteCA values in Table 1 are cluster-level isochrone-fit parameters. ~~The posterior distribution of the inferred parameters can be seen as a plot pair in Fig. B.4.~~

6.3. ~~Identification of YSOs~~

Young stellar objects (YSOs) are early-stage stars still accreting material from circumstellar disks; they include protostars and the more evolved PMS stars, the latter comprising classical T-Tauri and Herbig Ae/Be stars.

Our analysis aims to quantify the fraction of YSOs within the cluster, noted as $Y_{\rm frac}$, which serves as an age indicator. We determined the reddening-free parameter Q for each star using

$$Q = (J - H) - \frac{E(J-H)}{E(H-K)}(H - K), \tag{4}$$

where $E(J-H)/E(H-K)$ is assumed to be 1.55 (Mathis 1990). Following Buckner & Froebrich (2013), a star is considered a YSO if its Q value is less than -0.050 mag.

We followed the method of Cameron (2011) to estimate $Y_{\rm frac}$ and its uncertainty. Using a non-informative $\mathcal{B}(1,1)$ prior, ~~the~~ where $\mathcal{B}(a,b)$ denotes the Beta distribution with shape parameters a and b, the posterior distribution for $Y_{\rm frac}$ is modeled as

$$Y_{\rm frac} = \frac{N_{\rm YSO}}{N_{\rm cl}}, \tag{5}$$

where $N_{\rm YSO}$ is the number of YSOs and $N_{\rm cl} = 193$ is the number of reference-sample sources with 2MASS JHK_s photometry, for which Q can be computed (out of the 254 sources in the reference sample). The 95% credible interval for $Y_{\rm frac}$ was then derived from the posterior Beta distribution.

We identified 53 YSOs, with a mean Q value of -0.528, a median of -0.389, and a standard deviation of 0.421 for sources below -0.050 mag.

The resulting $Y_{\rm frac}$ is 0.275, with a 95% credible interval of [0.217, 0.342]. This value is conditional on 2MASS detectability: the 61 reference sources without 2MASS counterparts are systematically fainter (median G = 18.1 versus 16.6; KS $p \sim 10^{-6}$), and the limiting assumptions that none or all of them are YSOs bound $Y_{\rm frac}$ to $[0.21, 0.45]$. The value is higher than those reported by Buckner & Froebrich (2013), who found only 18 of 397 cluster candidates with $Y_{\rm frac} > 0.100$ and no cluster with $Y_{\rm frac} > 0.200$; since the two samples are not selection-matched, we read this as an indicative, not quantitative, sign that NGC 6383 is younger or more active in recent star formation than most clusters in their study.

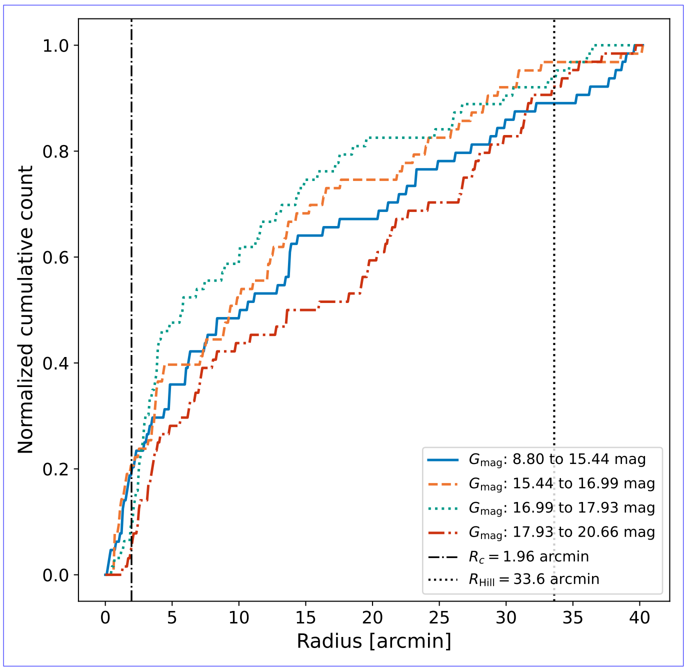

Fig. 8. Normalized cumulative counts of reference-sample stars ($\tilde{p} \gtrsim 0.6$) within NGC 6383 are plotted against radial distance from the cluster center for different brightness ranges, segmented into quartiles of G_{mag}: 8.80–15.44 (blue solid), 15.44–16.99 (orange dashed), 16.99–17.93 (teal dotted), and 17.93–20.66 mag (red dash-dotted). The black dash-dotted and black dotted vertical lines mark the core radius $R_c = 1.96$ arcmin and the Hill radius $R_{Hill} = 33.6$ arcmin; the adopted King outer radius, $R_t = 54^{+7}_{-11}$ arcmin (Appendix D), lies beyond the plotted range of this 40 arcmin sample.

6.3. ~~Luminosity, mass, and dynamical state~~

7. Luminosity, mass, and dynamical state

Having established the cluster's membership, structure, and stellar content, we now examine its luminosity function, individual stellar masses, and dynamical state, all derived for the $\tilde{p} \gtrsim 0.6$ reference sample. For NGC 6383, we converted apparent G magnitudes to absolute magnitudes using the parallax-derived distance. Figure C.2 shows the luminosity function (LF), indicating an increase toward dimmer magnitudes.

The radial cumulative distributions of cluster members at various magnitude levels, illustrated in Fig. 8, show that fainter stars, ranging from 17.93 to 20.66 mag, are less concentrated around the cluster's center. This observation is consistent with *Gaia*'s completeness limit for magnitudes between 18 and 19 at distances around 1 kpc (Boubert & Everall 2020; Castro-Ginard et al. 2023). ~~A~~ To test whether this reflects a real spatial difference, we compared the radial distribution of the faintest quartile with that of each of the three brighter quartiles using two-sample Kolmogorov-Smirnov (KS) ~~test gives $p = 0.010$ and 0.052 against~~ tests. The faintest quartile differs from the two intermediate quartiles ~~but~~ ($G = 16.99$–17.93 mag: $D = 0.28$, $p = 0.010$; $G = 15.44$–16.99 mag: $D = 0.23$, $p = 0.052$), but is statistically indistinguishable from the brightest quartile ($G = 8.80$–15.44 mag: $D = 0.16$, $p = 0.42$~~against the brightest one; only the first comparison survives~~); after a Holm correction ~~over~~ for the three tests, ~~and we read the pattern as a completeness effect rather than a dynamical one~~only the difference with respect to the 16.99–17.93 mag quartile remains significant. If the lower central concentration of the faintest stars were caused by dynamical mass segregation, the contrast should be strongest with respect to the brightest (most massive) quartile, which is the opposite of what we find. We therefore attribute the apparent deficit of faint stars toward the crowded cluster center to *Gaia* incompleteness at $G \gtrsim 18$ mag rather than to a dynamical effect.

We estimated individual stellar masses and binary probabilities with ASteCA v0.6.9 (Fig. C.1). These values, particularly for PMS stars, should be interpreted with caution because the PMS CMD is poorly defined (Da Rio et al. 2010), which limits isochrone-based mass and binarity estimates. Binary probabilities of at least 0.6 define the binary class used throughout, consistent with the membership thresholds, while secondary masses are added to the system mass only for binary probability > 0.7, a stricter cut that avoids inflating system masses through uncertain binary identifications. ~~The~~

Both the primary mass m_1 and the secondary mass m_2 are individual per-star estimates returned by ASteCA, not the product of an assumed universal mass ratio. For each of the 200 synthetic-cluster realizations drawn from the posterior of the fitted parameters, every observed star is matched to its photometrically nearest synthetic star, whose primary and (when the match is a binary) secondary masses are recorded; m_1 and m_2 are the medians of these values over the realizations, and the binary probability is the fraction of realizations in which the match is a binary. The mass ratios of the synthetic binaries follow the Duchêne & Kraus (2013) distribution introduced in Sect. 6.1; for the 51 stars whose secondary mass is added to the system mass (binary probability > 0.7), the resulting mass ratios span $q = m_2/m_1 = 0.26$–0.73 with a median of 0.60. The average mass of the stars is $1.31 \pm 0.10\,M_\odot$ (standard error on the mean, combining the sampling and per-star mass uncertainties), computed as the total mass per system (summing both components for binary probability > 0.7) over the 254-member reference sample.

To investigate mass segregation, we constructed radial normalized cumulative mass distributions for single stars, binary stars, and the combination of both, as shown in Fig. 9. A two-sample KS test comparing binary and single stars yields $D = 0.221$ and $p = 0.010$, indicating that binary stars are significantly more centrally concentrated. The signal is stable against the classification threshold ($D = 0.21$–0.29, $p \leq 0.018$ for binary-probability cuts of 0.5–0.8) and is not solely a mass effect: restricting both classes to their overlapping mass range (0.37–2.77 $M_\odot$) gives $D = 0.19$, $p = 0.07$, and a nearest-mass-matched comparison gives $D = 0.36$, $p < 0.001$, although mass and binarity remain partially entangled. Across single-star mass quartiles the KS results vary ($D = 0.12$–0.33, $p = 0.03$–0.86); given the width of these bins and the spread in p-values, we do not rely on any single quartile, and instead confirm the segregation with the bin-free Λ_{MSR} analysis (Fig. 10) below.

To contextualize these findings, we calculated the half-mass relaxation time t_{rh} (Spitzer 1969), which is crucial for estimating the minimum mass segregation time t_{seg}. The half-mass relaxation time is given by:

$$t_{rh} = \cancel{\frac{0.17N}{\ln(\lambda N)}} \frac{0.138\,N}{\ln(\gamma N)} \sqrt{\frac{r_{hm}^3}{GM}} \qquad (6)$$

where $N = 254$ is the size of the reference sample, ~~$\lambda = 0.110$ is a constant~~ $\gamma = 0.110$ is the equal-mass Coulomb-logarithm constant (not to be confused with the HDBSCAN density level λ

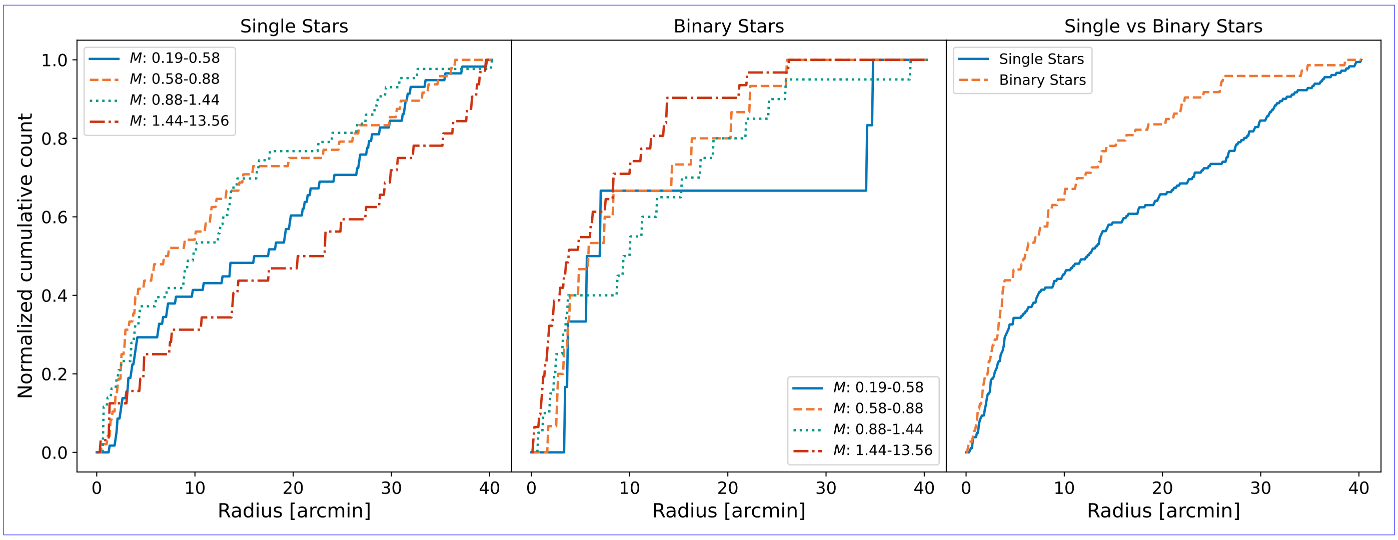

Fig. 9. *Left panel:* The cumulative distributions of single stars within NGC 6383 are segmented into four mass quartiles ranging from 0.19–0.58 $M_\odot$ to 1.44–13.56 $M_\odot$, illustrating the spatial distribution across different mass ranges and highlighting the potential influences of mass on the distribution of stars within the cluster. *Middle panel:* The same distributions as in the left panel for binary stars only, where binaries are defined as sources with a binary probability of at least 0.6. *Right panel:* The normalized cumulative distributions compare single stars with binary stars, with masses between 0.19 $M_\odot$ and 13.56 $M_\odot$. In all panels the quartiles are drawn as blue solid, orange dashed, teal dotted, and red dash-dotted curves; in the right panel single stars are the blue solid and binaries the orange dashed curve. All plots are based on stars with a membership pseudo-probability of at least 0.6. The corresponding Kolmogorov-Smirnov (KS) test results are given in Sect. 7.

of Sect. 3.2; Giersz & Heggie 1994), and $r_{\rm hm} = 2.02 \pm 0.41$ pc is the half-mass radius, derived from the radial cumulative mass distribution. For self-consistency between N and M, we take M as the total observed mass of the same stellar system, $M = N\langle m\rangle = 332 \pm 26\,M_\odot$, rather than an IMF-extrapolated total cluster mass. The calculated half-mass relaxation time for NGC 6383 is ~~30.5 ± 9.4 Myr.~~ 24.7 ± 7.6 Myr (Spitzer & Hart 1971). For a system with a mass spectrum the effective Coulomb constant is smaller ($\gamma \simeq 0.02$; Giersz & Heggie 1996), which would roughly double $t_{\rm rh}$; since all our dynamical-youth arguments rely on the cluster age being much shorter than $t_{\rm rh}$, adopting the equal-mass value is the conservative choice.

Using the half-mass relaxation time, ~~we can estimate $t_{\rm seg}$ for the cluster's most massive star by following the approach of~~ the timescale on which two-body relaxation drives a star of mass m toward the cluster center can be estimated following Spitzer (1969):

$$t_{\rm seg}(m) = \frac{\langle m\rangle}{m}\, t_{\rm rh}, \tag{7}$$

where $\langle m\rangle$ is the average ~~mass of a star~~ stellar mass in the cluster~~, and m is the mass of~~ ; the more massive the star, the faster it segregates. For the most massive star ~~. This results in a~~ of the reference sample (13.56 ± 3.25 $M_\odot$) this yields the minimum segregation time of ~~2.94 ± 1.17 Myr~~ the cluster, $t_{\rm seg} = 2.38 \pm 1.24$ Myr (propagating the uncertainties in $\langle m\rangle$, m, and $t_{\rm rh}$)~~for the~~ ; since this is shorter than the 3.53 Myr cluster age, the most massive star ~~of 13.56 ± 3.25 $M_\odot$, indicating it has had sufficient time to undergo mass segregation.~~ has had time to segregate dynamically. This value depends directly on the mass of the most massive member and would decrease if HD 159176, which we do not retain as a member (Sect. 8.1.1), belonged to the cluster. Adopting the component masses of $\simeq$ 44–46 $M_\odot$ derived by Penny et al. (2016), $t_{\rm seg}$ of the most massive object would drop to $\simeq$ 0.7 Myr ($\simeq$ 0.4 Myr if the binary is treated as a single $\sim$ 90 $M_\odot$ body). This is a factor of 3–7 shorter and would only reinforce this conclusion.

~~We evaluated the potential for primordial mass segregation by comparing the radial distributions of stars with a~~ For all other stars, Eq. (7) implies progressively longer segregation times. Setting $t_{\rm seg}(m)$ equal to the cluster age defines a limiting mass $m_{\rm lim} = \langle m\rangle\, t_{\rm rh}/3.53\,{\rm Myr} = 9.1\,M_\odot$ (computed with unrounded posterior values). Stars less massive than this have $t_{\rm seg}$ ~~greater than 3.53 Myr, which corresponds to the age of the cluster. The sample includes stars with masses ranging from 0.19 $M_\odot$ to~~ longer than the cluster age, so two-body relaxation cannot yet have segregated them: any central concentration they exhibit must be primordial or early-dynamical. If HD 159176 were included as a member, $m_{\rm lim}$ would shift only to $\simeq 10.3\,M_\odot$, leaving this subsample unchanged. We therefore repeated the analysis of Fig. 9 for the subsample with $m < m_{\rm lim}$. This subsample comprises 252 of the 254 reference stars (99.2%) and spans 0.19–8.49 $M_\odot$~~, which corresponds to 99.2% of the reference sample~~; only the two most massive members, of 13.13 and 13.56 $M_\odot$, are excluded. The results ~~, shown in Fig.C.5, revealed a similar behavior to Fig.9, where the highest mass bin has a distinct distribution. This similarity is expected , given that the sample is nearly identical to the one used in the previous analysis~~(Fig. C.5) closely resemble Fig. 9, as expected for a nearly identical sample. Within this subsample the single-star quartile KS results range over D = 0.12–0.36 and p = 0.01–0.84, with the upper mass quartile (1.41–8.49 $M_\odot$) showing the largest separation. In this sample, the average mass is 1.21 ± 0.08 $M_\odot$. The distributions of single and binary stars are similar up to approximately 5 arcmin, beyond which single stars show less concentration compared to binary stars.

As an independent, bin-free check of mass segregation, we computed the mass-segregation ratio $\Lambda_{\rm MSR}$ of Allison et al. (2009b), which compares the minimum-spanning-tree (MST) length of the $N_{\rm MST}$ most massive members with that of random sets of the same size. This estimator needs no cluster center, and for smooth, centrally concentrated regions it agrees with the other standard mass-segregation diagnostics (Parker & Goodwin 2015). We verified that NGC 6383 belongs to this regime rather than being spatially substructured by computing the structure parameter $\mathcal{Q}$ of Cartwright & Whitworth (2004) (the ratio of the normalized mean MST edge length to the normalized mean projected separation, not to be confused with the reddening-free index Q used above to identify YSOs). We obtain $\mathcal{Q} = 1.03 \pm 0.03$ (bootstrap over the 254 members), well above the $\mathcal{Q} \simeq 0.8$ boundary separating centrally concentrated from fractal, substructured distributions, and consistent with the King concentration ~~$C = 1.32$~~$C = 1.43$; we note that a proper-motion-selected, parallax-clipped sample is itself biased toward smooth configurations, so $\mathcal{Q}$ serves here as a consistency check rather than an independent morphology measurement. As shown in Fig. 10, Λ_{MSR} reaches ≈ 2.5–3.6 for the 5–15 most massive members (up to $\sim 3\sigma$ above unity) and decreases toward unity for larger samples, confirming that the most massive members are significantly more centrally concentrated than typical members. The signal is unchanged if the median or geometric mean of the random MST lengths replaces the arithmetic mean, a variant that is robust against single massive outliers (Maschberger & Clarke 2011). This bin-independent result supports the mass-segregation signal seen in the cumulative distributions.

The cluster is dynamically young, with an age of only ~~$\simeq 0.12\,t_{rh}$~~$\simeq 0.14\,t_{rh}$, far below the $\sim$ 2–3.5 t_{rh} over which primordial mass segregation is erased in N-body models, including those with a realistic primordial binary population (Pavlík 2020). The intermediate-mass binaries that are centrally concentrated have segregation times longer than the cluster age, so two-body relaxation is too slow to explain their distribution, suggesting instead that they formed closer to the center than single stars.

Two-body relaxation is not the only dynamical route, however: cool or substructured clusters can segregate rapidly through the transient dense core formed during an early collapse (Allison et al. 2009a), and subcluster merging accelerates segregation even in the low-membership regime of local open clusters such as NGC 6383 (Moeckel & Bonnell 2009). Such early-dynamical segregation also erases substructure, leaving a smooth, concentrated remnant like the one observed here ($\mathcal{Q} = 1.03$); it is therefore observationally degenerate with genuine birth segregation. We describe the central concentration as primordial or near-primordial, where near-primordial encompasses rapid early-dynamical segregation during cluster assembly. Simulations show that this configuration can be inherited if the cluster assembled from clumps that were already mass-segregated, or segregated before merging (McMillan et al. 2007), and can be amplified by residual-gas expulsion, which preferentially removes peripheral low-mass stars (Baumgardt & Kroupa 2007). A comparable primordial or near-primordial segregation is seen in other young clusters of similar age (Chen et al. 2007; Sabbi et al. 2008).

Population-level studies caution against a purely primordial reading for young open clusters: applying Λ_{MSR} and related indicators to 3881 *Gaia* DR3 clusters, Zhang (2026) finds the spatial segregation of < 10 Myr systems to be statistically consistent with random and attributes classical segregation to two-body relaxation only beyond $\sim$ 100 Myr, and N-body models compared with observed open clusters similarly find primordial segregation not to be a decisive ingredient (Amiri et al. 2026). Our single-cluster detection therefore does not by itself settle the primordial-versus-dynamical question; we report the segregation as measured and its origin as primordial or early-dynamical. ~~Simulations show that this configuration can be inherited if the cluster assembled from clumps that were~~

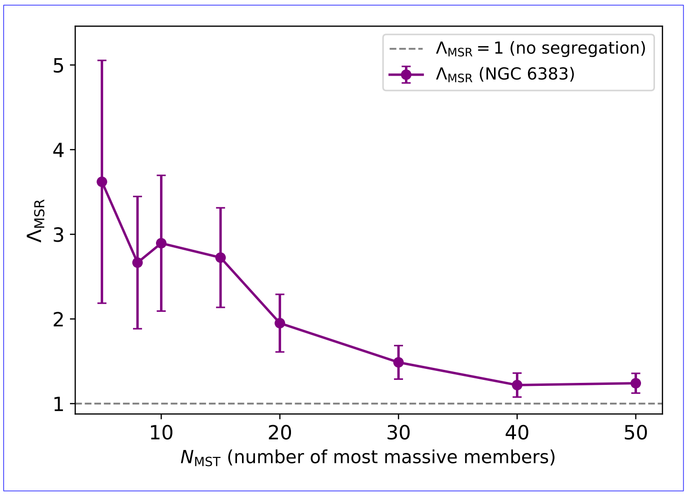

Fig. 10. Mass-segregation ratio Λ_{MSR} (Allison et al. 2009b) of NGC 6383 as a function of the number of most-massive members N_{MST}. Following Allison et al. (2009b), $\Lambda_{MSR} = \langle l_{norm} \rangle / l_{massive}$ is the ratio of the mean minimum-spanning-tree length of 500 random N_{MST}-star sets to that of the N_{MST} most massive members (purple circles connected by a solid line); error bars are $\sigma_{norm}/l_{massive}$. Values $\Lambda_{MSR} > 1$ indicate that the most massive stars are more centrally concentrated than random members; the gray dashed line marks $\Lambda_{MSR} = 1$ (no segregation).

~~already mass-segregated, or segregated before merging , and can be amplified by residual-gas expulsion, which preferentially removes peripheral low-mass stars . A comparable primordial or near-primordial segregation is seen in other young clusters of similar age .~~

8. Comparison with previous studies

8.1. ~~Comparative analysis with cluster members and PMS study~~HD 159176

8.1.1. ~~HD 159176~~

HD 159176 is a double-lined O-type spectroscopic binary projected close to the center of NGC 6383 and is responsible for ionizing the surrounding HII region (Stickland et al. 1993; Linder et al. 2007; Penny et al. 2016). The estimated age of HD 159176 is ~~2.30~~2.3–~~2.80 Myr~~ 2.7 Myr (Rauw et al. 2010)~~. This is compatible with the~~, in agreement with the earlier estimate of 2.8 ± 0.5 Myr (Fitzgerald et al. 1978). This estimate places the binary components, with the parameters of Linder et al. (2007), on single-star evolutionary tracks with and without rotation (Meynet & Maeder 2003), while the 2–3 Myr ~~age of the most likely low-mass PMS candidates found by PMS~~ ages of Rauw et al. (2010) ~~,~~ rest on comparing X-ray-selected candidates with several PMS model families (e.g. Siess et al. 2000), and are therefore subject to the same code-dependent systematics discussed in Sect. 6.1. The two age ranges are compatible, consistent with a common star-formation event if the binary is assumed to be associated with the ~~same star-formation event~~cluster.

In the study by Aidelman et al. (2018), it was hypothesized that if HD 159176 were a blue straggler, it would imply an older age for NGC 6383, ranging between 6.00 and 10.0 Myr. We do not use this assumption in the present analysis. HD 159176 is classified in the spectroscopic literature as ~~an~~ a double-lined O-type main-sequence binary, ~~and its~~ O7V((f))+O7V((f)) (Stickland et al. 1993; Linder et al.

2007) or O6.5V+O7V (Penny et al. 2016), and its thermal X-ray emission is ~~naturally associated with massive-star winds and wind interaction rather than with an accreting Be-binary scenario~~ attributed to the winds of the two O stars and their wind–wind interaction (Rauw et al. 2003; De Becker et al. 2004); no accretion-powered scenario has been proposed for this system.

The *Gaia* DR3 values of HD 159176 are $\varpi = 1.167 \pm 0.071$ mas, $\mu_{\alpha*} = 2.621 \pm 0.083$, and $\mu_\delta = -0.798 \pm 0.058$ mas yr^{-1}. These must be treated cautiously: at $G \simeq 5.7$ (vs $G_{\rm min} = 8.80$ and median $G \simeq 17.0$ for the reference sample) the star lies in the bright-source regime prone to saturation and magnitude-dependent astrometric systematics (Lindegren et al. 2021b,a; Cantat-Gaudin & Brandt 2021; Maíz Apellániz 2022). The $\mu_{\alpha*}$ agrees with the cluster mean and the parallax offset is not by itself conclusive; the strongest tension is in μ_δ, which differs from the cluster mean (-1.713 mas yr^{-1}, dispersion 0.138) by $\Delta\mu_\delta \simeq 0.91$ mas yr^{-1}, about 6σ once the star's own proper-motion uncertainty is added in quadrature to the member dispersion. We note that the preprocessing parallax window (0.750–1.10 mas) already excludes the DR3 parallax of HD 159176, so its absence from the candidate sample is partly by construction; given the astrometric-quality caveats above, we read the μ_δ tension as supporting, rather than decisive, evidence against membership.

Gaia DR3 astrometric-quality indicators reinforce this: HD 159176 has a normal RUWE = 0.937, 13 visibility periods, and no duplicated-source flag, but an excess-noise significance of 135.6 and an IPD multi-peak fraction of 39, far above the reference-sample medians of zero (84th percentiles 1.53 and 0). These are not used as a membership criterion but confirm that HD 159176 must be treated as a special bright-source astrometric case (Rybizki et al. 2022).

We therefore exclude HD 159176 from the member sample used to characterize NGC ~~6383. Its~~ 6383, its noteworthy projected central position ~~remains noteworthy, but the *Gaia* DR3 proper motion, especially in declination, argues against treating it as a secure member in the present analysis~~notwithstanding. The age quoted for NGC 6383 in this paper is derived from the selected low-mass and PMS population and does not rely on the evolutionary status of HD 159176.

8.1.1. ~~NGC 6383 22~~

8.2. NGC 6383 22

NGC 6383 22[9] ~~,~~ is identified as a λ Boo star. These stars are chemically peculiar A-type stars with abundance anomalies attributed to the accretion of metal-poor material. In a spectroscopic survey of metal-weak and emission-line stars in the Southern Hemisphere, Murphy et al. (2020) found that λ Boo stars, including NGC 6383 22, exhibit excesses at longer wavelengths, suggestive of circumstellar disks. This characteristic aligns with other stars in NGC 6383 and those studied in Kalari (2019). However, NGC 6383 22 is not retained in our *Gaia*-selected member sample because, despite matching the position and parallax of the cluster, its proper motion ($\mu_{\alpha*} = 1.790 \pm 0.030$ mas yr^{-1}, $\mu_\delta = -2.710 \pm 0.024$ mas yr^{-1}) is more than 3σ away from the mean of the posterior distribution of μ. This result indicates that this star is likely passing near or through the cluster.

8.2.1. ~~PMS studies using photometry and VPHAS+~~

8.3. PMS studies using Hα photometry and VPHAS+

Kalari (2019) examined 55 classical T-Tauri stars (CTTS) in the star-forming region Sh 2-012 and NGC 6383, utilizing ~~optical~~ VPHAS+ optical and Hα photometry and *Gaia* astrometry. They reported a median age of 2.80 ± 1.60 Myr for these stars, with masses ranging from 0.3 to 1.0 $M_\odot$. Their ages and masses were obtained by interpolating the observed r versus $r - i$ CMD positions against PARSEC evolutionary models reddened by $E(B - V) = 0.32$ (Bressan et al. 2012); Kalari (2019) notes that the same data yield mean ages of 2.9 and 3.5 Myr with the Tognelli et al. (2011) and Siess et al. (2000) models, so part of any difference from our MIST-based age reflects the choice of evolutionary code rather than the data (Sect. 6.1). Notably, 94% of CTTS with near-infrared cross-matches conform to the near-infrared T-Tauri locus, and all stars with mid-infrared photometry showed signs of accreting circumstellar disks; the CTTS are concentrated around NGC ~~6383.~~ 6383 and toward the bright rims of Sh 2-012.

Out of the 55 CTTS kinematic members of Sh 2-012, only 15 are cataloged as members or potential members of the cluster in this study, with most having a PMS probability over 0.6[10]. The difference in member selection is mainly methodological: Kalari (2019) selected CTTS primarily from photometric Hα equivalent widths and then refined membership with a double-peaked Gaussian fit to the *Gaia* proper motions, whereas our selection rests on proper-motion clustering. Only one T-Tauri source[11] is likely a binary, with a 0.94 probability of being a binary star according to our analysis; this source belongs to both our $\tilde{p} \geq 0.6$ reference sample and the Kalari (2019) CTTS list.

We also cross-matched our 321 likely candidates with the UBVRI Hα photometric catalog of Rauw et al. (2010). Using a 1.0 arcsec matching radius, we found 141 counterparts, including 124 sources from the $\tilde{p} \geq 0.6$ reference sample. Their CDS table provides the R_C−Hα index but not a published source-by-source Hα emitter flag. We therefore do not treat the match as a recovery of the original Rauw et al. emitter list. Instead, we used their published R_C−Hα color offsets, 0.12–0.24 mag above the main-sequence R_C−Hα relation for candidates and > 0.24 mag for emitters, as a diagnostic relative to an empirical non-emitter locus derived from their catalog. Under this diagnostic classification, 51 of the 141 matched sources show Hα excesses, including 28 of the 57 matched Sagitta PMS candidates. For the reference sample alone, 23 of the 51 matched Sagitta PMS candidates show such an excess. These matches support the presence of an emission-line subset among the PMS candidates, but the absence of diagnostic Hα excess for the remaining PMS candidates is not decisive because Rauw et al. (2010) emphasized that Hα emission is variable, can be weak for part of the PMS population, and that Hα-selected samples can be contaminated by foreground and background objects.

~~Out of the 55 CTTS kinematic members of Sh 2-012, only 15 are cataloged as members or potential members of the cluster in this study, with most having a PMS probability over 0.6[12]. The difference in member selection is mainly methodological: selected CTTS primarily from photometric Hα equivalent widths and then refined membership with a double-peaked Gaussian fit to the *Gaia* proper motions, whereas our selection rests on~~

[9] Cataloged in SIMBAD as Gaia DR3 ~~4054615634716139264~~4054615634716139264.

[10] The four exceptions have PMS probabilities of 16, 12, 55, and 45%.

[11] Cataloged as Gaia DR3 4054567805963940352.

[12] ~~The four exceptions have PMS probabilities of 16, 12, 55, and 45%.~~

~~proper-motion clustering. Only one T-Tauri source~~[12] ~~is likely a binary, with a 0.94 probability of being a binary star according to our analysis; this source belongs to both our $\tilde{p} \geq 0.6$ reference sample and the CTTS list.~~

8.3.1. ~~Comparison with published catalogs~~

8.4. Comparison with published catalogs

We compared our likely-candidate catalog (see also the historical compilation in Appendix A) with the catalogs from Cantat-Gaudin et al. (2020), Jaehnig et al. (2021), He et al. (2022), Hunt & Reffert (2024), and the members listed in SIMBAD. We find 148 sources in common with Cantat-Gaudin et al. (2020), 97 present only in their catalog, and 173 only in ours; 161 in common with Jaehnig et al. (2021) (123 only theirs, 160 only ours); 90 with He et al. (2022) (47 only theirs, 231 only ours); 202 with Hunt & Reffert (2024) (120 only theirs, 119 only ours); and 168 with SIMBAD (163 only there, 153 only ours). Because all rely at least partly on *Gaia* astrometry, the differences mainly reflect methodological choices, spatial windows, probability thresholds, and whether PMS or photometric information enters the selection. In particular, our selection clusters in the proper-motion plane only, whereas most catalog pipelines cluster in the full three- to five-dimensional astrometric space with their own probability floors and search radii, and SIMBAD aggregates heterogeneous literature memberships.

These overlaps also delimit the added value of the present analysis. These survey catalogs are homogeneous all-sky compilations optimized for thousands of clusters: for any individual cluster they provide a member list and bulk parameters derived from a single, fixed pipeline configuration. For NGC 6383, this work adds four elements. (i) Posterior distributions, with reported uncertainties, for the mean astrometric parameters, the distance, the King structural parameters, and the isochrone age, rather than pipeline point estimates. (ii) An explicit sensitivity budget for the main configuration choices, which none of the above catalogs quantify on a per-cluster basis: the HDBSCAN hyperparameter sweep is marginalized into the per-star pseudo-probabilities (Sect. 3.2, Appendix B), the dependence of the membership and structural parameters on the search window is measured directly (Appendix D), and the mass-segregation signal is checked for stability against the binary-classification threshold (Sect. 7). (iii) Cluster-specific analyses that survey pipelines do not attempt, namely the PMS census (Sagitta probabilities, ages, and extinctions; the YSO fraction; and the Hα cross-match with Rauw et al. 2010), the dynamical-state analysis ($t_{\rm rh}$, $t_{\rm seg}$, $\Lambda_{\rm MSR}$), per-star masses and binary probabilities, and the individual treatment of HD 159176 as a bright-source astrometric special case. (iv) A per-star catalog at the CDS (Table 2) together with the open-source pipeline, which make the selection reproducible and its parameter dependence auditable.

Where a derived quantity does depend on a configuration choice (most notably the outer membership and the King outer radius with respect to the search window) we flag it explicitly as window- or model-dependent rather than reporting it as a robust measurement (Sect. 5, Appendix D). The survey catalogs remain the appropriate reference for homogeneous population-level comparisons across clusters; we regard the two approaches as complementary rather than competing.

The adopted King outer radius also merits comparison with the literature. Published outer or tidal radii for NGC 6383 span 15–30 arcmin (15.0, Kharchenko et al. 2005; 29.0 ± 4.2, Piskunov et al. 2008; 22.7, Hunt & Reffert 2024; 29.7 ± 7.7, Pulgar-Escobar & Henríquez-Salgado 2024; Table A.1), all smaller than our $R_t = 54^{+7}_{-11}$ arcmin, although the largest of them is compatible with it at the $\sim 2\sigma$ level. The difference is expected in direction: those determinations were limited either by extractions comparable to or smaller than the radii they report or by the depth of the underlying survey data, so, like our own 40 arcmin fit ($R_t = 40^{+16}_{-17}$ arcmin), their outer radii are limited by the available field, whereas our adopted value comes from the only extraction with a fitted background annulus beyond the cluster (Appendix D). The comparison therefore mostly reflects the window and background treatment, reinforcing our reading of R_t as a model- and prior-dependent outer scale rather than a sharp physical boundary.

9. Conclusions

This study presents an analysis of the young open cluster NGC 6383, employing Bayesian analysis and machine-learning techniques to identify cluster members and determine key parameters. Using the HDBSCAN algorithm in proper-motion space followed by parallax clipping, we identified 321 likely cluster candidates with $\tilde{p} > 0.5$. The reference sample used for the cluster parameters contains 254 sources with membership pseudo-probabilities of at least 60%.

Our findings indicate that the cluster has a mode age of $3.53^{+1.40}_{-1.00}$ Myr, with a distance of 1.11 ± 0.06 kpc derived from the parallax-based distance estimation. The ~~fitted core and King outer radii are 1.95 ± 0.19 arcmin and 40.4 ± 14.3 arcmin, respectively~~core radius is $1.96^{+0.19}_{-0.16}$ arcmin; the King outer radius, measured on the 70 arcmin extraction that fully encloses the fitted profile, is 54^{+7}_{-11} arcmin. Wider-field tests with 50, 60, and 70 arcmin input cones recover an NGC 6383-like proper-motion branch and a consistent outer radius in all cases, but the selected outer membership is not invariant with field size (~~the per-radius parameters are compiled in~~ Table D.2, Appendix D)~~. In~~; in line with *Gaia*-based work on open-cluster structure and tidal features (Angelo et al. 2025; Risbud et al. 2025; Xu et al. 2025), ~~this outer radius~~ R_t should therefore be regarded as a model-dependent scale rather than as an independently measured physical boundary. ~~We also observed mass segregation among the binary stars; given the cluster's dynamical youth, two-body relaxation is too slow to have produced it, pointing instead to a primordial or early-dynamical origin.~~

The CMD analysis revealed a well-defined main sequence and a population of PMS stars, with an apparent age spread of 1.6–6.3 Myr, consistent with a recent and possibly extended star-formation episode. The Sagitta neural network provided additional insights into the probabilities of PMS stars, their ages, and the extinction affecting cluster members. A cross-match with the UBVRI Hα catalog of Rauw et al. (2010) identifies a diagnostic Hα-excess subset among the matched PMS candidates, consistent with an emission-line or accreting component.

~~Our analysis aligns with previous studies, offering updated parameters for NGC 6383.~~ HD 159176 is projected near the cluster center, but its *Gaia* DR3 proper motion in declination is inconsistent with the selected cluster population~~. HD 159176 lies well outside the magnitude range of the *Gaia*-selected candidates, so its *Gaia* astrometry must be read with caution; even so, the proper-motion~~ ; even allowing for the bright-source

[12] ~~Cataloged as Gaia DR3 4054567805963940352~~

astrometric caveats discussed in Sect. 8.1.1, the tension is large enough that we exclude it from the member sample used for the cluster characterization. The age estimate reported here is derived from the selected PMS and low-mass population, as in previous PMS-based studies, and does not depend on assuming any particular evolutionary state for HD 159176.

The central concentration of binaries, even for those with segregation times longer than the cluster's age, suggests that they formed closer to the cluster center, pointing to a primordial or early-dynamical origin for the segregation. ~~This~~ As discussed in Sect. 7, this reading is consistent with simulations of ~~segregation inheritance through~~ clump merging and ~~of gas expulsion , and with primordial~~ gas expulsion and with the segregation observed in other young clusters, although population-level analyses ~~argue that segregation in young open clusters is largely dynamical~~ find little statistically significant spatial segregation in clusters younger than ~ 10 Myr (Zhang 2026); our per-cluster result does not resolve this debate.

This work combines machine-learning membership with Bayesian parameter estimation to deliver an updated characterization of NGC 6383, and provides a template for similar studies of open-cluster candidates in the *Gaia* era. The software used in this study is available on GitHub[12], and the catalog is available at the CDS (see Data availability).

Data availability

The full 321-source likely-member catalog (Table 2) is available in electronic form at the CDS via anonymous ftp to `cdsarc.cds.unistra.fr` (`130.79.128.5`) or via `https://cdsarc.cds.unistra.fr/viz-bin/cat/J/A+A/`. The COSMIC software used for this analysis is available at `https://github.com/notluquis/COSMIC`.

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[12] https://github.com/notluquis/COSMIC
[13] `http://www.astropy.org`

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Appendix A: Historical parameters of NGC 6383

Table A.1 compiles the structural and fundamental parameters of NGC 6383 reported in the literature since Trumpler (1930). The wide spread of historical core and tidal radii, distances, reddenings, distance moduli, and ages~~—~~, obtained with heterogeneous data and methods over nearly a century~~—~~, illustrates the lack of a homogeneous determination and motivates the *Gaia*-based reanalysis presented in this work. For each study we list only the quantities explicitly reported; ellipses denote parameters not provided.

Appendix B: HDBSCAN diagnostic and ASteCA posterior corner plots

This appendix documents the unsupervised-clustering diagnostics and the isochrone-fit posterior that underpin the membership and parameter determinations of ~~Sect.3~~Sects. 3 and 6.1. We first describe how the HDBSCAN configuration was selected from the proper-motion clustering sweep (Figs. B.1 and B.2), show the resulting clustering of the full sample in the proper-motion plane and on the sky (Fig. B.3), and then present the joint posterior of the ASteCA isochrone fit (Fig. B.4).

HDBSCAN was run over the grid $m_{\rm cl} = 10, \dots, 299$ in the two-dimensional proper-motion plane, setting `min_cluster_size=min_samples=`$m_{\rm cl}$ and using the `leaf` cluster-selection method (Sect. 3.2). For each run we recorded the size of the recovered NGC 6383 branch ~~and its persistence in the condensed tree, retaining only the runs that pass the branch-stability criteria (maximum~~ (the condensed-tree ~~level~~ branch containing the highest density level $\lambda_{\rm max}$; Sect. 3.2) and its $\lambda_{\rm max}$. Figure B.1 displays the runs in the stable regime, delimited by $\lambda_{\rm max} \geq 8$ and a recovered branch of 200–701 sources ~~). Figure B.1 showsthat~~ (the upper bound being the largest branch recovered in the sweep); below $\lambda_{\rm max} = 8$ the condensed tree is too shallow to separate the cluster from the field. These display cuts do not enter the selection: as Fig. B.1 shows, the recovered branch size peaks at the adopted $m_{\rm cl} = 43$~~, the value adopted throughout:~~, with smaller $m_{\rm cl}$ fragmenting the cluster into spurious sub-groups ~~, whereas~~ and larger $m_{\rm cl}$ progressively ~~merge~~ merging field stars into the branch or ~~dissolve~~ dissolving it altogether.

The condensed cluster tree (Fig. B.2) displays the HDBSCAN hierarchy as a function of the density level $\lambda = 1/d_{\rm mreach}$, where wider and more persistent branches correspond to denser, more significant over-densities. The NGC 6383 branch ~~—~~(the tall blue ellipse on the left~~—~~) persists over the widest λ interval and is clearly detached from the field branches on the right, providing the qualitative counterpart of the quantitative stability criterion used in Fig. B.1. This persistence is what allows the proper-motion over-density of NGC 6383 to be selected unambiguously, despite the known tendency of HDBSCAN to report spurious dense fluctuations in noisy data (Hunt & Reffert 2021).

Figure B.3 places the selected branch in context, showing the full 40 arcmin preprocessed sample (15276 sources) labeled by HDBSCAN, both in the proper-motion plane and on the sky. In proper motion, NGC 6383 forms a compact, well-isolated over-density around $(\mu_\alpha^*, \mu_\delta) = (2.5, -1.7)$ mas yr^{-1}, clearly separated from the diffuse field and from the other, more dispersed HDBSCAN branches. On the sky, by contrast, the three groups overlap, with only a mild central concentration of the cluster members. This confirms that the membership signal is carried by the proper motions rather than by projected position, justifying the proper-motion-only clustering feature space adopted in Sect. 3.2.

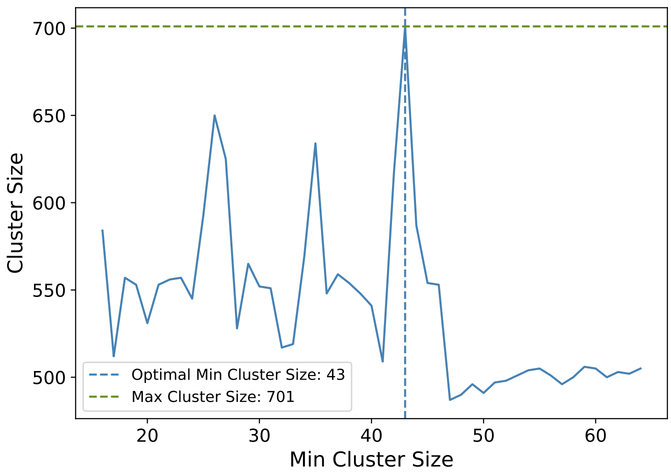

Fig. B.1. Cluster size variation as a function of minimum cluster size, determined through HDBSCAN clustering on the proper-motion data of the sources. The blue line ~~represents~~ shows the recovered cluster ~~sizes achieved at various minimum cluster sizes. The~~ size; the dashed blue line ~~highlights~~ marks the ~~optimal~~ adopted minimum cluster size of 43, where the cluster size ~~reaches a peak. The~~ peaks, and the dashed green line ~~indicates~~ the maximum observed cluster size of 701. Only sweep runs ~~passing~~ in the ~~branch-stability criteria~~ stable regime (~~condensed-tree~~ $\lambda_{\rm max} \geq 8$ and recovered branch ~~size between 200 and 701~~ of 200–701 sources) are shown, which is why the axis does not span the full $m_{\rm cl}$ = 10–299 grid; this display cut does not enter the selection of $m_{\rm cl}$.

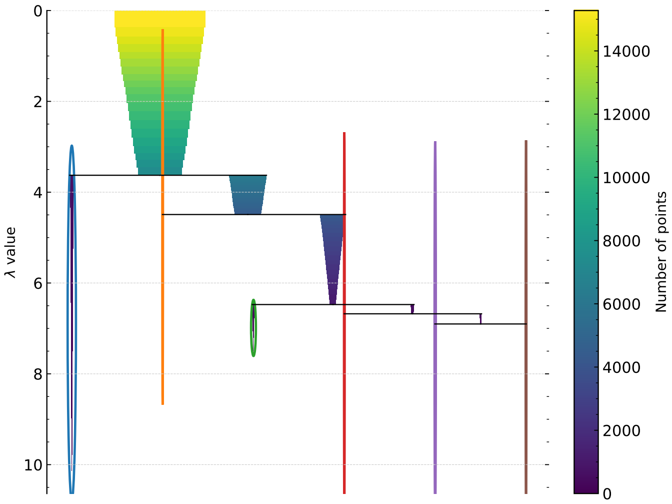

Fig. B.2. Condensed cluster tree (dendrogram) from the HDBSCAN analysis: NGC 6383 cluster sources on the left, HDBSCAN-identified field sources on the right. Branch width and color give the number of sources at each level (color bar); $\lambda = 1/d_{\rm mreach}$ is the inverse mutual-reachability distance. Colored ellipses mark the leaf clusters selected by HDBSCAN; the tall blue ellipse on the left, persisting over the widest λ range, is the branch adopted as NGC 6383.

Figure B.4 shows the joint and marginal posteriors of the four ASteCA isochrone-fit parameters together with the likelihood-scatter nuisance term σ, sampled with the DEMetropolis ensemble (Sect.~~3~~ 6.1). All marginals are single-peaked and well sampled. The off-diagonal panels make the classical CMD-fitting degeneracies explicit: logarithmic age correlates with both visual extinction and distance modulus, so that an older age can be partially compensated by a lower extinction or a smaller dis-

Table A.1. Historical review of observed parameters for the open cluster NGC 6383, gathered from various studies. The table details the number of cluster members, core and tidal radii, parallax, distance, E(B-V) color excess, distance modulus ($m-M$), and estimated ages. The literature tidal radii are King tidal radii, here denoted the King outer radius R_t, except Kharchenko et al. (2005), whose value is their empirical cluster (corona) radius; the Piskunov et al. (2008) value is their r_t (8.3 ± 1.2 pc at their adopted distance). This summary provides a view on how measurements and interpretations of NGC 6383 have evolved over time.

Reference	Members	Core Radius arcmin	Tidal Radius arcmin	Parallax mas	Distance kpc	E(B-V) mag	$(m-M)$ mag	Age Myr
Trumpler (1930)	...	...	...	...	2.13	...	...	...
Sanford (1949)	...	...	...	...	0.76	...	...	...
Eggen (1961)	...	...	...	...	...	0.30	...	...
Graham (1967)	74	...	...	...	1.38	...	10.7±0.5	...
Evans (1968)	At least 21	...	...	...	1.30	0.35	10.6	5.0
Lindoff (1968)	...	...	...	...	1.25	...	10.5	∼ 20
Becker & Fenkart (1971)	...	...	...	...	1.07	0.26	10.9	...
Fitzgerald et al. (1978)	...	1.25	...	...	1.50±0.20	0.33±0.02	10.9	1.7±0.4
Lloyd Evans (1978)	...	...	...	...	1.35	0.35	10.6	Up to 5.0
The et al. (1985)	...	...	...	...	1.40±0.15	0.30±0.01	...	...
Pandey et al. (1989)	...	...	...	...	...	0.35	11.7	∼ 4.5
Battinelli & Capuzzo-Dolcetta (1991)	27	...	...	...	1.38	0.34	...	∼ 20
Feinstein (1994)	...	...	...	...	1.40	0.33	...	...
Rauw et al. (2003)	...	...	...	...	...	0.33	10.7	...
Kharchenko et al. (2005)	13	4.80	15.0	...	0.99	0.30	10.9	5.0
Paunzen et al. (2007)	...	...	...	...	1.70±0.30	0.29±0.05	...	Less than 4.0
Piskunov et al. (2008)	...	...	29.0±4.2	...	0.99	0.30	10.9	5.0
Rauw et al. (2010)	55 PMS candidates	...	...	...	...	0.32	...	2–3
Aidelman et al. (2018)	...	...	...	...	0.83±0.16	0.51±0.03	9.61±0.38	∼3–10
Kalari (2019)	At least 55	...	...	...	...	...	...	2.8±1.6
Jaehnig et al. (2021)	284	...	...	0.93±0.09	1.07	...	...	...
He et al. (2022)	...	...	...	0.88±0.05	...	0.45	...	3.5
Hunt & Reffert (2024)	322	2.49	22.7	0.89±0.08	1.10	0.38	10.2	∼ 4.0
Pulgar-Escobar & Henríquez-Salgado (2024)	266	1.21±0.13	29.7±7.7	0.92±0.06	1.10±0.04	0.47±0.03	10.9±0.33	∼1–4

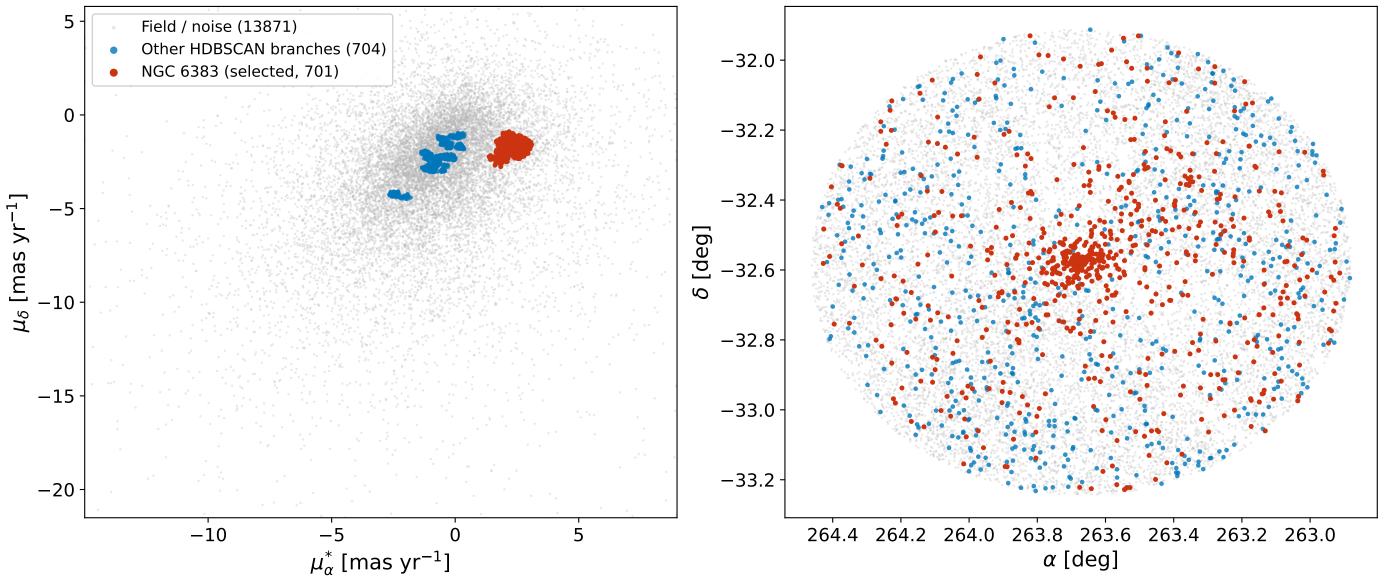

Fig. B.3. Full HDBSCAN clustering panorama of the 40 arcmin preprocessed sample (15276 sources). *Left:* proper-motion plane (μ_α^*, μ_δ); *Right:* on-sky distribution in right ascension and declination. Sources are color-coded by their HDBSCAN label: the selected NGC 6383 branch (red), the other HDBSCAN branches (blue), and the field/noise (gray); the legend lists the number of sources in each group. The proper-motion axes are clipped to the 0.5–99.5 percentile range so that the cluster over-density is visible against the broad field distribution. NGC 6383 is a compact, isolated clump in proper motion but overlaps the field on the sky, illustrating that the membership separation is driven by proper motion.

tance. Metallicity is the least constrained parameter and remains close to prior-dominated, consistent with the age–metallicity–extinction degeneracy noted in the main text. The credible intervals quoted in the main text and in Table 1 correspond to these marginal posteriors; σ is retained only as a fit nuisance parameter and carries no physical meaning.

Appendix C: Supplementary figures

This appendix collects the supplementary figures referenced from the main text: the ~~on-sky distribution of the members with the derived scale radii overlaid on the DSS2-red image (Fig. C.3); the~~ masses and binarity CMD (Fig. C.1); the luminosity function derived from absolute magnitudes (Fig. C.2); the on-sky distribution of the members with the derived scale radii overlaid on the DSS2-red image (Fig. C.3); the per-star Sagitta PMS-probability, age, and extinction distributions (Fig. C.4); the

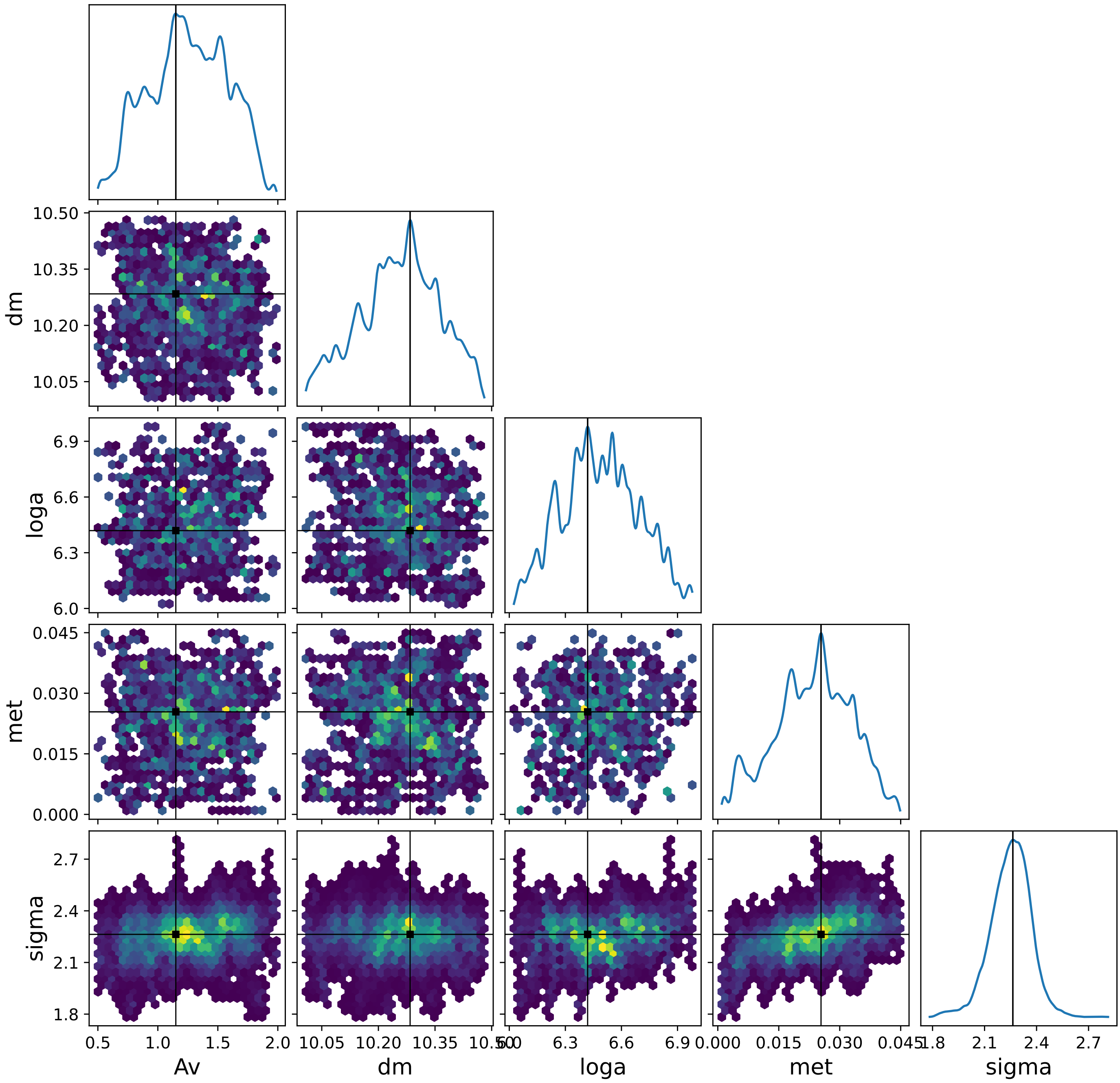

Fig. B.4. Posterior distributions of the parameters A_V (visual extinction), *dm* (distance modulus), *loga* (logarithmic age), *met* (metallicity), and the nuisance likelihood-scatter parameter σ. The σ parameter is the likelihood-scatter term used in the PyMC diagnostic implementation that generated this figure and should not be interpreted as a physical cluster dispersion. The diagonals display the marginal distributions for each parameter (blue curves). The off-diagonal hexbin plots illustrate the joint distributions, highlighting correlations between parameters. Black lines mark the mode of each distribution in all panels; in the off-diagonal panels, the joint mode is additionally marked by a black square.

mass-segregation cumulative distributions for the segregation-time-restricted sample (Fig. C.5); and the *Gaia* DR3 radial velocities and their amplitudes (Fig. C.6). These figures support, but are not essential to, the results presented in the main text.

Appendix D: Search-window sensitivity of the derived parameters

This appendix complements the membership-recovery counts of Table D.1 by listing, side by side, the astrometric and structural parameters recovered at each input cone-search radius. Only the membership, astrometric, and King-structural quantities were recomputed for the wider fields; the photometric quantities (ages, masses, and the mass-segregation diagnostics) remain tied to the 40 arcmin reference catalog and are not repeated per radius. As discussed in Sect.~~3.3~~ 3.3, the catalog-level astrometry is stable across radii, whereas the structural parameters and the outer-membership counts drift once the automated branch selection becomes field-window dependent (at 60 and 70 arcmin)~~, reinforcing the search-window-sensitive interpretation of any quantity tied to the cluster boundary.~~ .

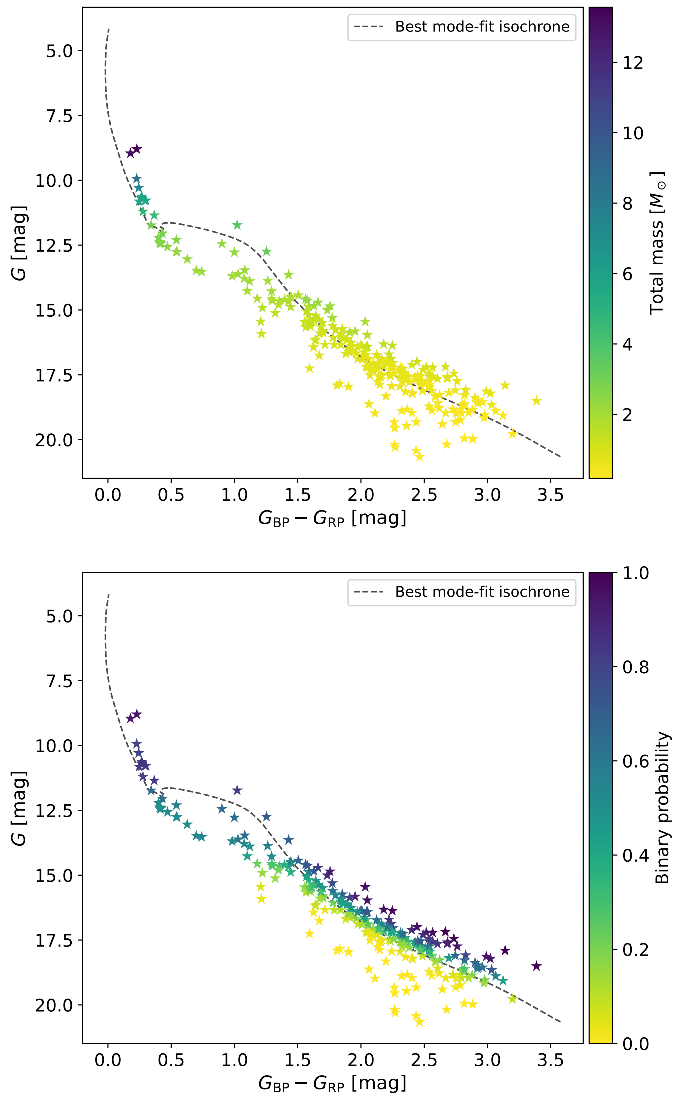

Fig. C.1. *Upper panel:* CMD displaying probable members and members of NGC 6383, color-coded by total mass, ranging from lower (yellow) to higher mass (purple). The dark-gray dashed line represents the best mode-fit isochrone, illustrating the evolutionary track inferred for the cluster. The total mass is calculated as the sum of the two components if the binary probability is greater than 0.7. *Lower panel:* The same stars as in the upper panel, color-coded by binary probability, from low (yellow) to high (purple). The dark-gray dashed line again represents the best mode-fit isochrone.

The 70 arcmin window provides the adopted King outer radius (Sect. 5). The probability that R_t exceeds the extraction window decreases from 51% at 40 arcmin to 40% and 13% at 50 and 60 arcmin (at 70 arcmin it vanishes by construction, since the prior support ends at $1.5\,T_{max} = 63.7$ arcmin), and Fig. D.1 shows that only this window reaches a well-defined background annulus beyond the cluster, with fitted background $b = 0.020^{+0.006}_{-0.010}$ and central density $k = 8.42 \pm 0.17$ stars per square arcminute. All R_t posteriors, including the adopted one, are truncated from above by the physically motivated $1.5\,T_{max}$ prior bound and remain conditional on it; the shift under a relaxed bound is quantified in Sect. 5. The background of this membership-selected sample measures the residual contamination level of the selection, not the raw field density.

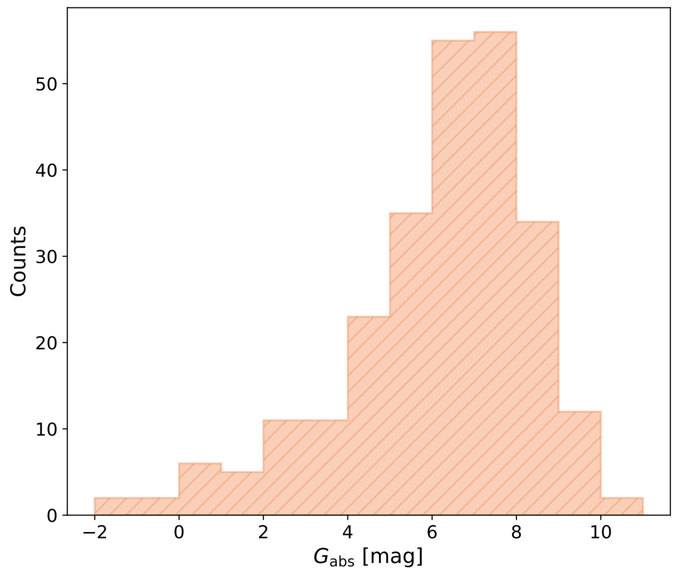

Fig. C.2. Luminosity function of NGC 6383: histogram (orange, hatched) of absolute magnitudes (G_{abs}) for stars with a membership pseudo-probability of at least 0.6.

The core radius and central density of the 70 arcmin fit are contamination-biased (Sect. 3.3) and are not adopted.

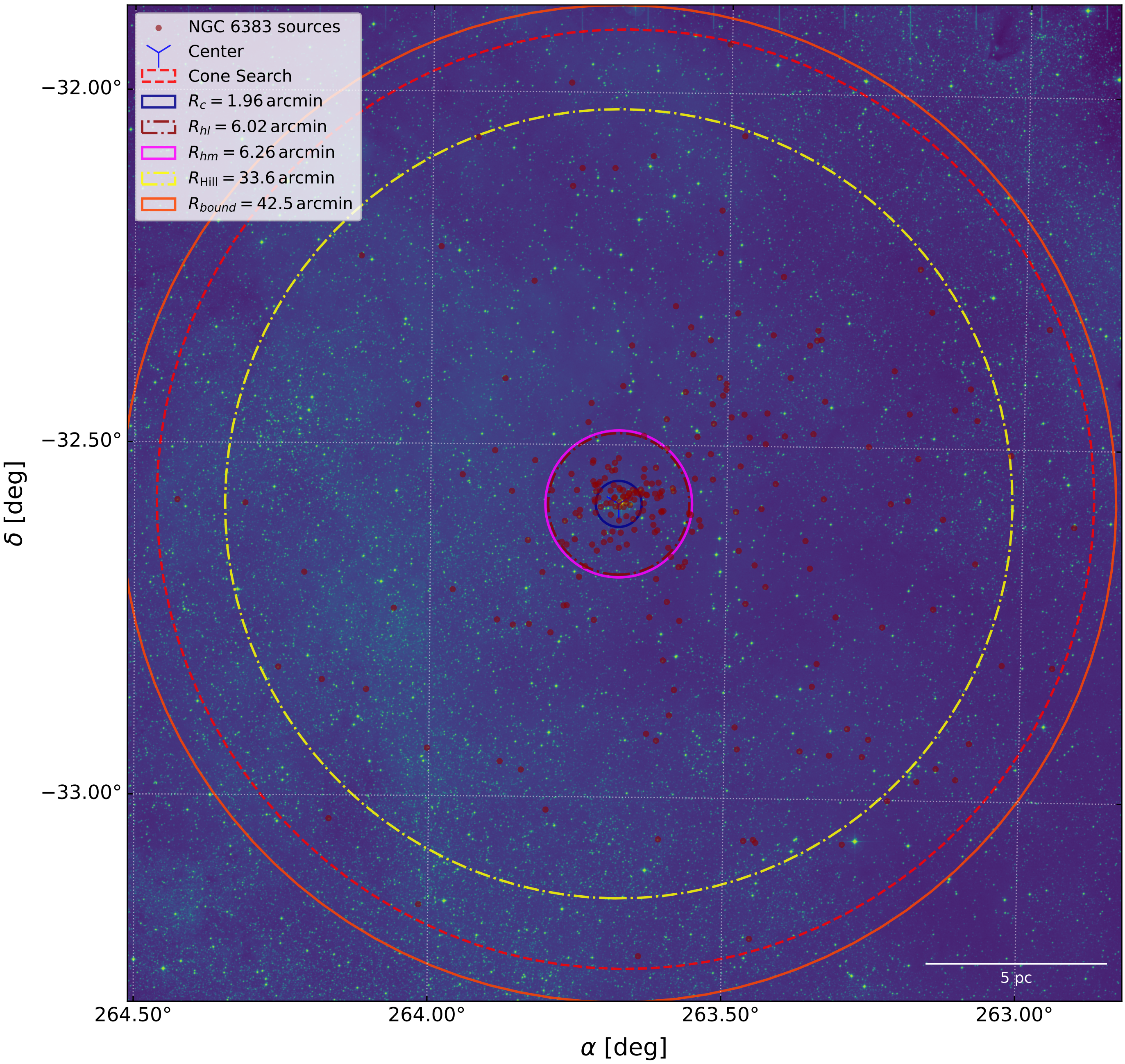

Fig. C.3. Spatial distribution of probable members and members (dark-red dots) overlaid on a DSS2-red background image (shown in false color), centered on the derived cluster center marked by a blue marker; the red dashed circle is the 40 arcmin cone-search radius. The image shows several fitted or derived scale radii: the ~~King outer radius R_t is depicted with a dash-dotted dark-slate-gray line, the~~ core radius R_c with a solid dark-blue line, the half-light radius R_{hl} with a dash-dotted burgundy line, the half-mass radius R_{hm} with a solid magenta line, the Hill radius ~~R_{hill}~~ R_{Hill} with a dash-dotted yellow line, and the gravitational bound radius R_{bound} with a solid orange line. The adopted King outer radius (R_t = 54 arcmin, Appendix D, Fig. D.1) lies just outside the displayed field and is not drawn. A scale bar indicating 5 pc is positioned at the bottom right to provide spatial reference relative to the cluster scale.

Table D.1. Robustness of the membership recovery against the input cone-search radius. The 40 arcmin row is the production catalog used for the adopted cluster parameters, while the 50–70 arcmin rows are wider-field robustness checks. Each run used the same preprocessing, the same proper-motion-only HDBSCAN feature space, the same pseudo-probability construction as in Sect. 3.2, and the same parallax-mode sigma clipping. The algorithm-selected branch is the branch selected by the generic sweep rule, while the NGC-like branch is the final HDBSCAN label closest to the reference proper motion of NGC 6383, $(\mu_{\alpha*}, \mu_\delta) = (2.542, -1.713)$ mas yr^{-1}.

Radius (arcmin)	Preprocessed sources	Best $m_{\rm cl}$	Algorithm label	NGC-like label	NGC branch size	$\tilde{p} > 0.5$ after clip	$\tilde{p} \geq 0.6$ reference sample	Common with adopted 254
40	15276	43	0	0	701	321	254	254
50	23844	18	28	28	767	355	236	203
60	33309	143	1	0	696	476	443	242
70	44438	177	1	0	798	650	628	252

Table D.2. Astrometric and King-structural parameters recovered at each input cone-search radius. The 40 arcmin row is the adopted production fit; the 50–70 arcmin rows use the same King priors applied to the wider-field reference samples. N is the number of reference-sample candidates ($\tilde{p} \geq 0.6$). The mean parallax $\bar{\varpi}$ is quoted with the 1σ dispersion of the subsample with fractional parallax error below 0.1; the proper-motion means carry the 1σ member dispersions. The core radius R_c, King outer radius R_t, and concentration $C = \log_{10}(R_t/R_c)$ are the posterior medians with 16–84% credible intervals of the King fit; $P(R_t > W)$ is the posterior probability that R_t exceeds the extraction window. Ages, masses, and mass segregation were not recomputed per radius (see text).

Radius (arcmin)	N ($\tilde{p} \geq 0.6$)	$\bar{\varpi}$ (mas)	$\mu_{\alpha*}$ (mas yr^{-1})	μ_δ (mas yr^{-1})	R_c (arcmin)	R_t (arcmin)	C	$P(R_t > W)$ (%)
40	254	0.908 ± 0.046	2.542 ± 0.152	-1.713 ± 0.138	$1.96^{+0.19}_{-0.16}$	40^{+16}_{-17}	$1.32^{+0.16}_{-0.27}$	51
50	236	0.906 ± 0.044	2.544 ± 0.133	-1.711 ± 0.127	$1.90^{+0.15}_{-0.13}$	46^{+12}_{-16}	$1.39^{+0.11}_{-0.20}$	40
60	443	0.916 ± 0.051	2.455 ± 0.239	-1.693 ± 0.159	$1.36^{+0.10}_{-0.09}$	47^{+12}_{-16}	$1.54^{+0.10}_{-0.19}$	13
70	628	0.915 ± 0.057	2.383 ± 0.279	-1.680 ± 0.163	$1.38^{+0.04}_{-0.04}$	54^{+7}_{-11}	$1.59^{+0.06}_{-0.11}$	0

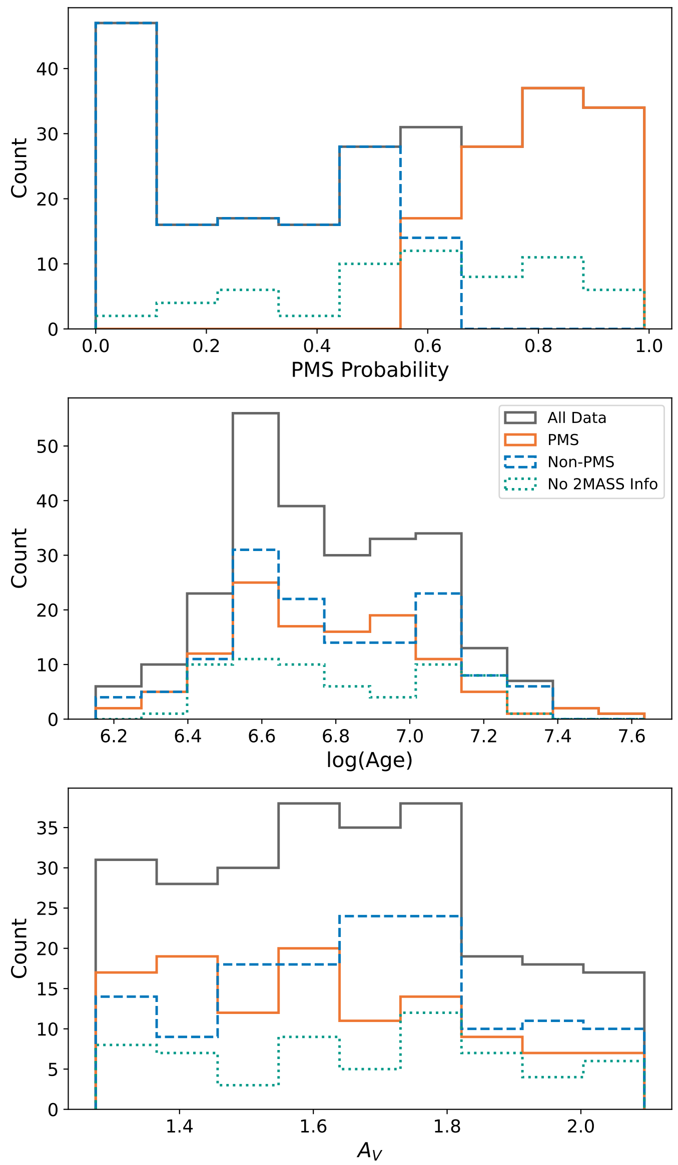

Fig. C.4. Histograms of the parameters obtained with Sagitta for the NGC 6383 reference sample ($\tilde{p} \geq 0.6$), including all data (gray solid), sources with a PMS probability of at least 60% (orange solid), sources with a PMS probability under 60% (blue dashed), and stars with missing 2MASS data (green dotted); the legend in the middle panel applies to all panels. The upper panel shows the distribution of PMS probability values, indicating the likelihood of stars being PMS; the middle panel presents the distribution of logarithmic age values, illustrating the range and frequency of stellar ages within the cluster; and the lower panel displays the distribution of visual extinction values (A_V), representing the amount of dust extinction affecting the observed stars.

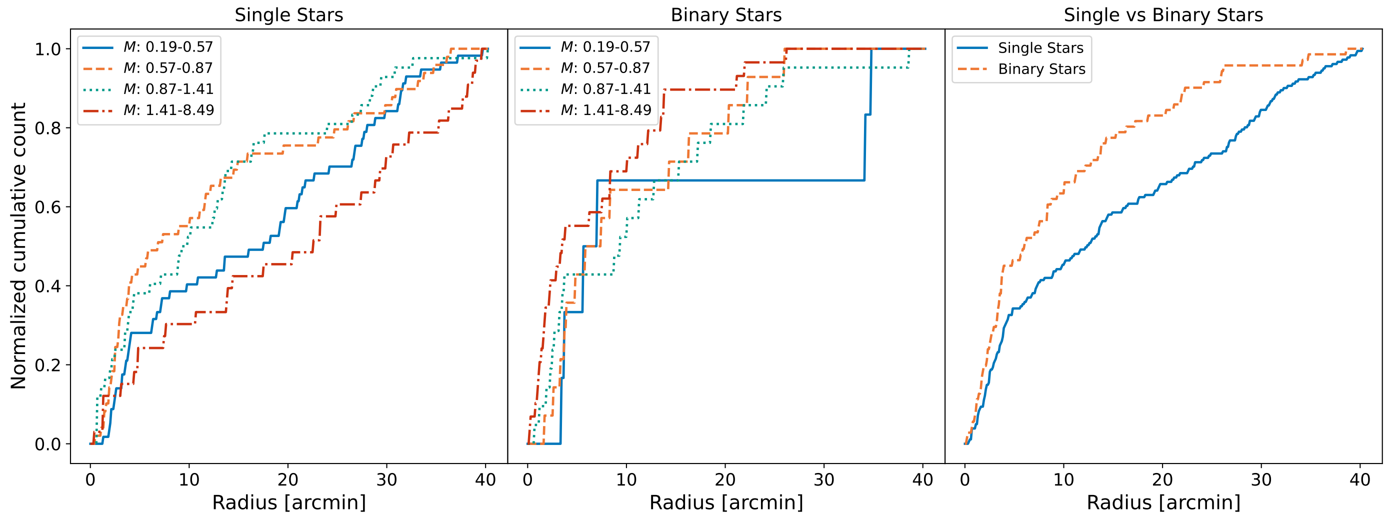

Fig. C.5. Same as Fig. 9, but ~~for~~ restricted to stars ~~with $M < 11.2\,M_\odot$ (~~whose segregation time ~~> 3.53 Myr,~~ exceeds the cluster age ($t_{\rm seg}(m) > 3.53$ Myr, i.e. $M < 9.1\,M_\odot$; Sect. 7); the mass quartiles are recomputed for this truncated sample (uppermost quartile 1.41–8.49 $M_\odot$).~~See Sect. 7 for a discussion.~~

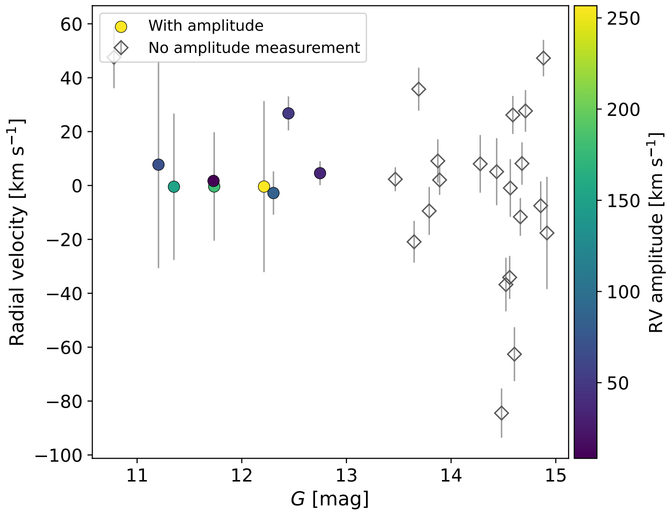

Fig. C.6. Radial velocity versus G magnitude for the 29 reference-sample sources with *Gaia* DR3 radial velocities. Filled circles have DR3 amplitude measurements (`rv_amplitude_robust`, which reflects the total amplitude in the radial-velocity time series following outlier removal) and are color-coded by amplitude (viridis color bar); open gray diamonds lack amplitude measurements. Vertical bars show the radial-velocity uncertainties.

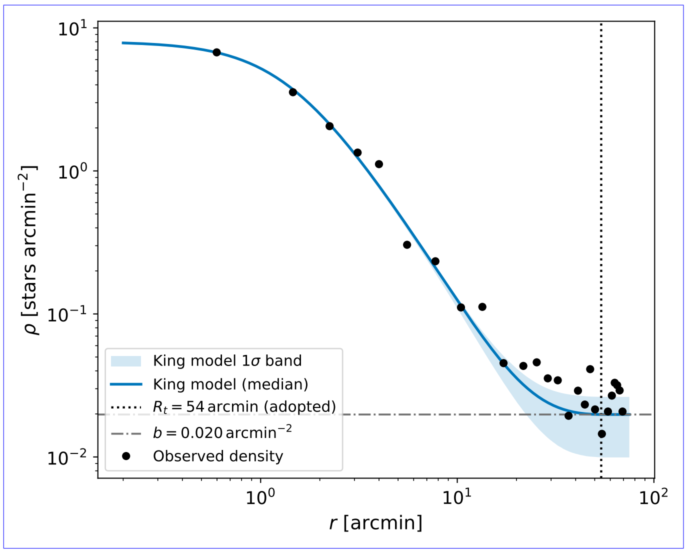

Fig. D.1. Radial density profile of the 70 arcmin extraction ($N = 628$, $\tilde{p} \geq 0.6$) with the King model at the posterior median (blue solid line) and its 16–84% posterior band (blue shading). The black dotted vertical line marks the adopted King outer radius ($R_t = 54$ arcmin) and the gray dash-dotted horizontal line the fitted background level b. Unlike the 40 arcmin fit (Fig. 5), this window contains a genuine background annulus beyond the cluster, so the outer radius is background-constrained; the higher central density and smaller core radius reflect the additional field contamination of this sample and are not adopted.